MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

REGISTER, PROGRAMMABLE, DIGITAL, KEY CODE FBH

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic programmable register module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 3.86 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FBH.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 70 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A20 and B21 through B40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/176-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

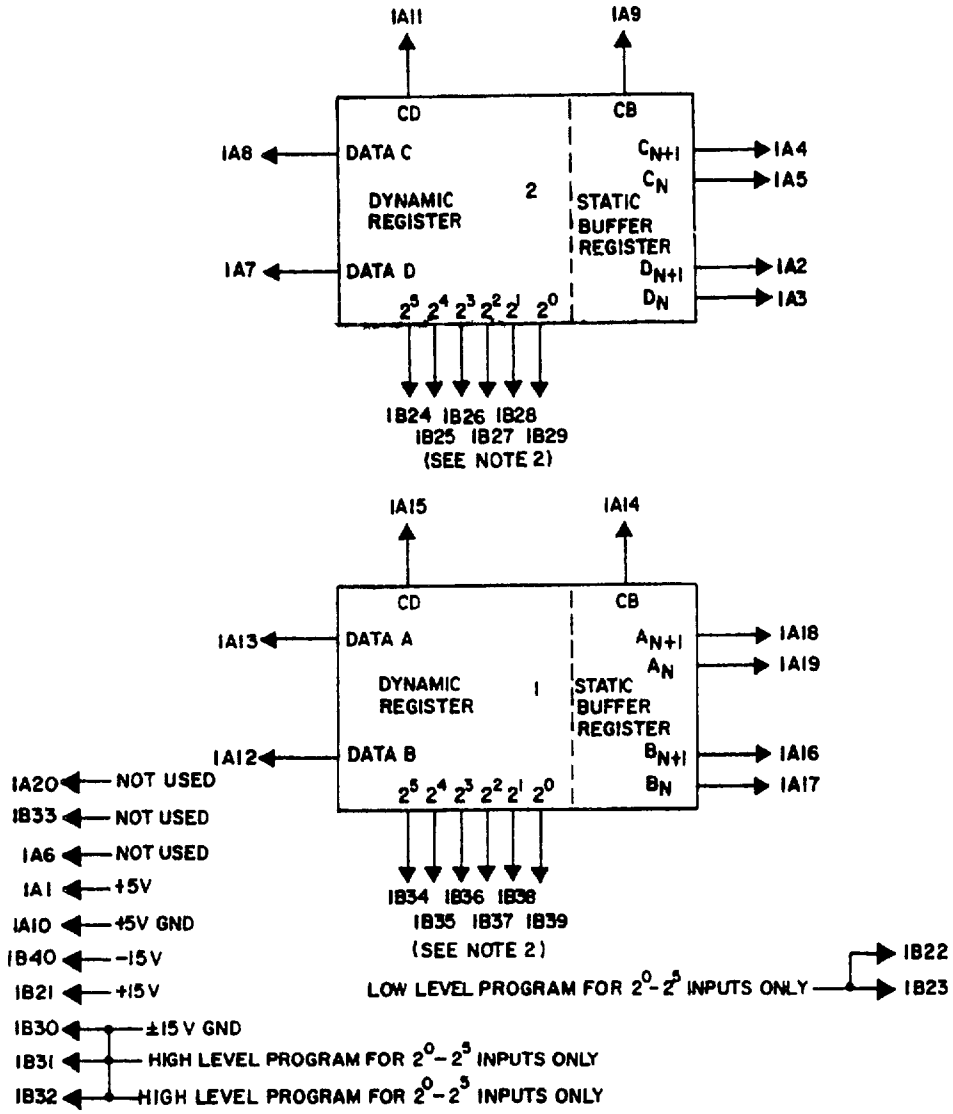
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. The 5 V and 15 V power grounds must be tied together externally.
2. Negative logic.

FIGURE 1. FBH module function diagram, register, programmable.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A8	V_{OUT} (low loaded)	III	S	0	800	0	800	0	800	0	800	V
B1-B8	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	mA
C1-C6	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1	Isolation, frame	VI	S	10	∞	1	∞	10	∞	1	∞	MΩ
D2-D5	Isolation, unused pins	VI	S	10	∞	1	∞	10	∞	1	∞	MΩ
E1-E8	Output voltage, low level	VII	X	0	400	0	400	0	400	0	400	V
E9-E16	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F	Program output voltage	VIII	X	9.5	14.5	9.5	14.5	9.5	14.5	9.5	14.5	V
G1-G8	Input current, low level	IX	X	0.10	1.73	0.10	1.73	0.10	1.73	0.10	1.73	mA
G9-G20	Input current, high level	IX	X	0	-100	0	-100	0	-100	0	-100	µA
H1-H8	Input current, high level	X	X	0	40	0	40	0	40	0	40	µA
H9-H20	Input current, low level	X	X	0	300	0	300	0	300	0	300	µA
J1-J8	Output short circuit current	XI	X	18	57	18	57	18	57	18	57	mA
J9	Output short circuit current, program	XI	X	19	24	19	24	19	24	19	24	mA
K1	Power supply current (5.5 V)	XII	S	5	150	5	150	5	150	5	150	mA
K2	Power supply current (+15.5 V)	XII	S	5	100	5	100	5	100	5	100	mA
K3	Power supply current (-15.5 V)	XII	S	0	30	0	30	0	30	0	30	mA
L1-L4	Maximum CD clock pulse width check	XIII	S	180	220	180	220	180	220	180	220	µs

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See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L5-L8	Input hold time check	XIII	S	1560	1640	1560	1640	1560	1640	1560	1640	ns
L9-L12	Input hold time check	XIII	S	1560	1640	1560	1640	1560	1640	1560	1640	ns
M1-M8	Transition time (t_{TLH})	XIV	S	4	50	4	50	4	50	4	50	ns
M9-M16	Transition time (t_{THL})	XIV	S	4	20	4	20	4	20	4	20	ns
N1-N8	Propagation delay time (t_{PLH})	XV	S	10	45	10	45	10	45	10	45	ns
N9-N12	Propagation delay time (t_{PLH})	XV	S	50	400	50	400	50	400	50	400	ns
N13-N20	Propagation delay time (t_{PLH})	XV	S	10	45	10	45	10	45	10	45	ns
P1-P8	Propagation delay time (t_{PHL})	XVI	S	10	45	10	45	10	45	10	45	ns
P9-P12	Propagation delay time (t_{PHL})	XVI	S	50	400	50	400	50	400	50	400	ns
P13-P20	Propagation delay time (t_{PHL})	XVI	S	10	45	10	45	10	45	10	45	ns
R1-R4	1 BIT Register operation	XVII	X	2	2	2	2	2	2	2	2	Count
R5-R8	2 BIT Register operation	XVII	X	3	3	3	3	3	3	3	3	Count
R9-R12	3 BIT Register operation	XVII	X	4	4	4	4	4	4	4	4	Count
R13-R16	5 BIT Register operation	XVII	X	6	6	6	6	6	6	6	6	Count
R17-R20	9 BIT Register operation	XVII	X	10	10	10	10	10	10	10	10	Count
R21-R24	17 BIT Register operation	XVII	X	18	18	18	18	18	18	18	18	Count
R25-R28	33 BIT Register operation	XVII	X	34	34	34	34	34	34	34	34	Count
R29-R32	64 BIT Register operation	XVII	X	65	65	65	65	65	65	65	65	Count
S1-S16	Output crosstalk, low level and high level	XVIII	Q	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
T1-T3	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	mV
U	Power supply filter capacitance	XX	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
V1-V20	Input capacitance	XXI	Q	2	15	2	15	2	15	2	15	pF
W1-W3	Continuity	XXII	X	0	1	0	1	0	1	0	1	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent of two modules from each inspection lot, whichever is greater) at 25° C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through U and W shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests E through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through W shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/176-1 supersedes part number M28787/176. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Power dissipation (clock pulse rate = 1.2 MHz)	
5.5 V supply	825 mW
+15.5 V supply	1550 mW
-15.5 V supply	465 mW
Supply voltage	5.0 ±0.5 V dc, ±15 ±0.5 V dc
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2 V minimum, V_{CC} maximum

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Input current (data and clock)		
Low level	(V_{IL} = 0 V)	1.73 mA maximum
	(V_{IL} = 0.4 V)	1.60 mA maximum
High level	(V_{IH} = 2.4 V)	40 µA maximum
	(V_{IH} = 5.5 V)	1 mA maximum
Output voltage		
Low level (I_{OL} = 16 mA)		0.4 V maximum
High level (I_{OH} = -800 µA)		2.4 V minimum
Data set up time		115 ns minimum for greater than two bit operation. 135 ns minimum for two bit operation. 150 ns minimum for one bit operation.
Data hold time		115 ns minimum
Clock triggering		Negative
CD clock pulse width		400 ns minimum, 50 µs maximum
CD clock pulse space		400 ns minimum, 50 µs maximum
CB clock pulse width		20 ns minimum
CB clock pulse space		20 ns minimum
CD to CB clock pulse delay		400 ns minimum
CB to CD clock pulse delay		0 ns minimum

6.5 Module function. This module contains two independent 2-bit dynamic shift registers with length variable from 1 to 64 bits and buffered output as specified in 6.5.1. The length of the dynamic register is determined by negative logic signals applied to the control lines 2^0, 2^1, 2^2, 2^3, 2^4, and 2^5. These lines are binary weighted by the value of the superscript and the length of the dynamic register is determined by adding one to the summation of the binary weighted values as illustrated in table II. Storage buffers provide simultaneous outputs for n and n+1 bits for each input. Separate clock inputs provide control for the programmable dynamic register (CD) and buffer (CB). CD and CB may be clocked simultaneously with the buffers then operating as additional register length.

6.5.1 Input to output relationship. Each register has the input to output relationship of figure 2. The shift register length control inputs (2^0-2^5) operate with negative logic. Refer to table II; connect the desired control pins to the program pin (1B22 or 1B23) for low level or connect to ground (1B30, 1B31, or 1B32) for high level.

6.5.2 Paralleling of outputs. Paralleling of outputs is not permissible.

6.6 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material except when being tested or in actual operation, to avoid buildup of static charge.

b. Soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

c. Signals should not be applied to the inputs while the device power supply is off.

d. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the protection input diodes.

e. All unused input leads must be connected to either V_{SS} (GND) or V_{DD} (device supply), whichever is appropriate, for the logic circuit involved.

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f. Before inserting modules into fixturing remove all power and input signals from the test fixture, then follow the sequencing below:

(1) Insert module into fixture.

(2) Apply power supplies in the following order: 5 V, 15 V, and -15 V.

(3) Apply input signals.

(4) Perform tests.

(5) Remove input signals.

(6) Remove power supplies in the following order: -15 V, 15 V, and 5 V.

(7) Remove module from fixture.

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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TABLE II. Dynamic register length. 1/-3/

2^5	2^4	2^3	2^2	2^1	2^0	Dynamic register length
L	L	L	L	L	L	1 bit
L	L	L	L	L	H	2 bits
L	L	L	L	H	L	3 bits
L	L	L	L	H	H	4 bits
L	L	L	H	L	L	5 bits
L	L	L	H	L	H	6 bits
L	L	L	H	H	L	7 bits
L	L	L	H	H	H	8 bits
L	L	H	L	L	L	9 bits
-	-	-	-	-	-	---
-	-	-	-	-	-	---
-	-	-	-	-	-	---
-	-	-	-	-	-	---
-	-	-	-	-	-	---
-	-	-	-	-	-	---
-	-	-	-	-	-	---
H	L	L	L	L	L	33 bits
-	-	-	-	-	-	---
-	-	-	-	-	-	---
-	-	-	-	-	-	---
-	-	-	-	-	-	---
-	-	-	-	-	-	---
-	-	-	-	-	-	---
H	H	H	H	H	H	64 bits

1/ L = Low-level (steady state).

2/ H = High-level (steady state).

3/ Negative logic.

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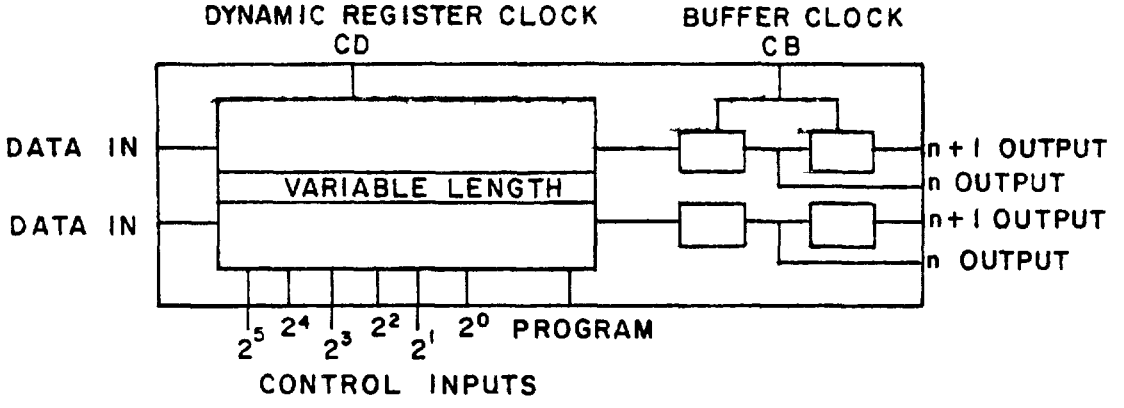
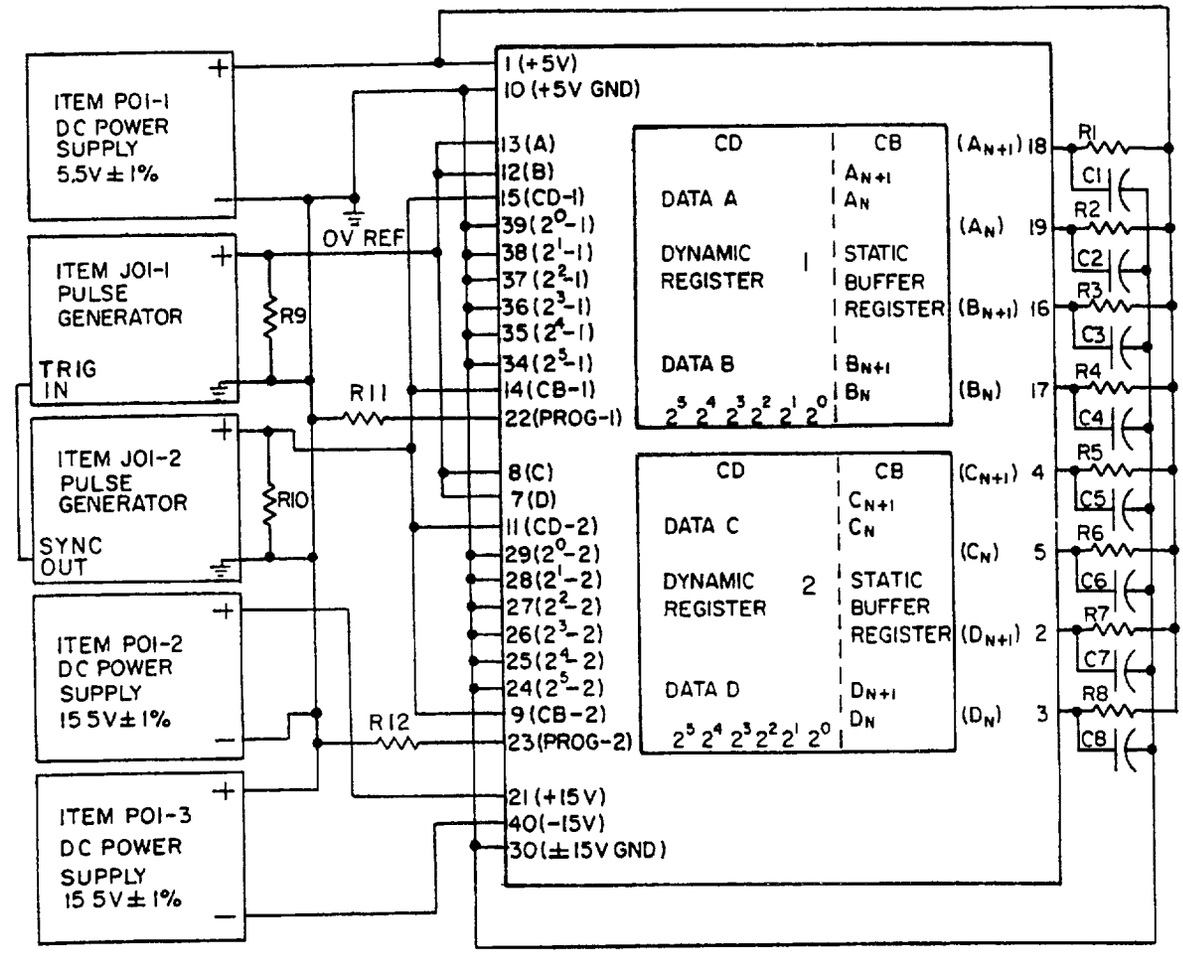


FIGURE 2 Input to output relationship.

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NOTES:
1. R1-R8 = 324Ω ±1%, 1/4 W.
2. R9 = R10 = 51Ω ±5%, 1 W.
3. R11 = R12 = 2.37 kΩ ±1%, 1/8 W.
4. C1-C8 = 100 pF ±20%.
5. Input pulse parameters:
   PWA = 2.0 ±0.2 µs.
   PWB = 1.0 ±0.1 µs.
   $t_{TLH} = t_{THL} = 50 \pm 25$ ns.
   $t_B = 500 \pm 100$ ns.
6. Refer to handling considerations in 6.6.

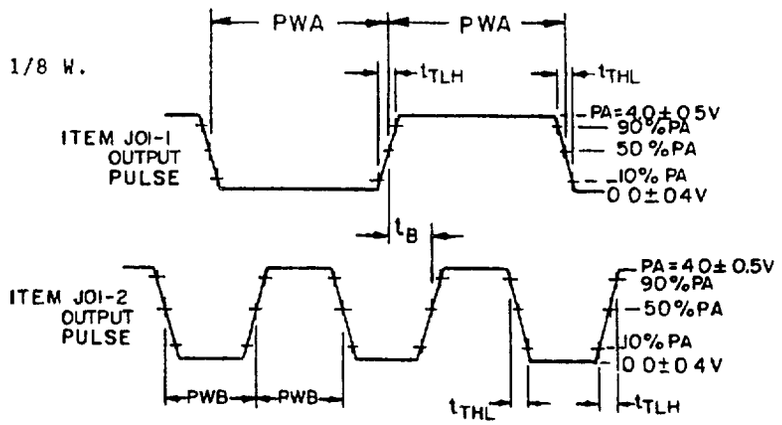
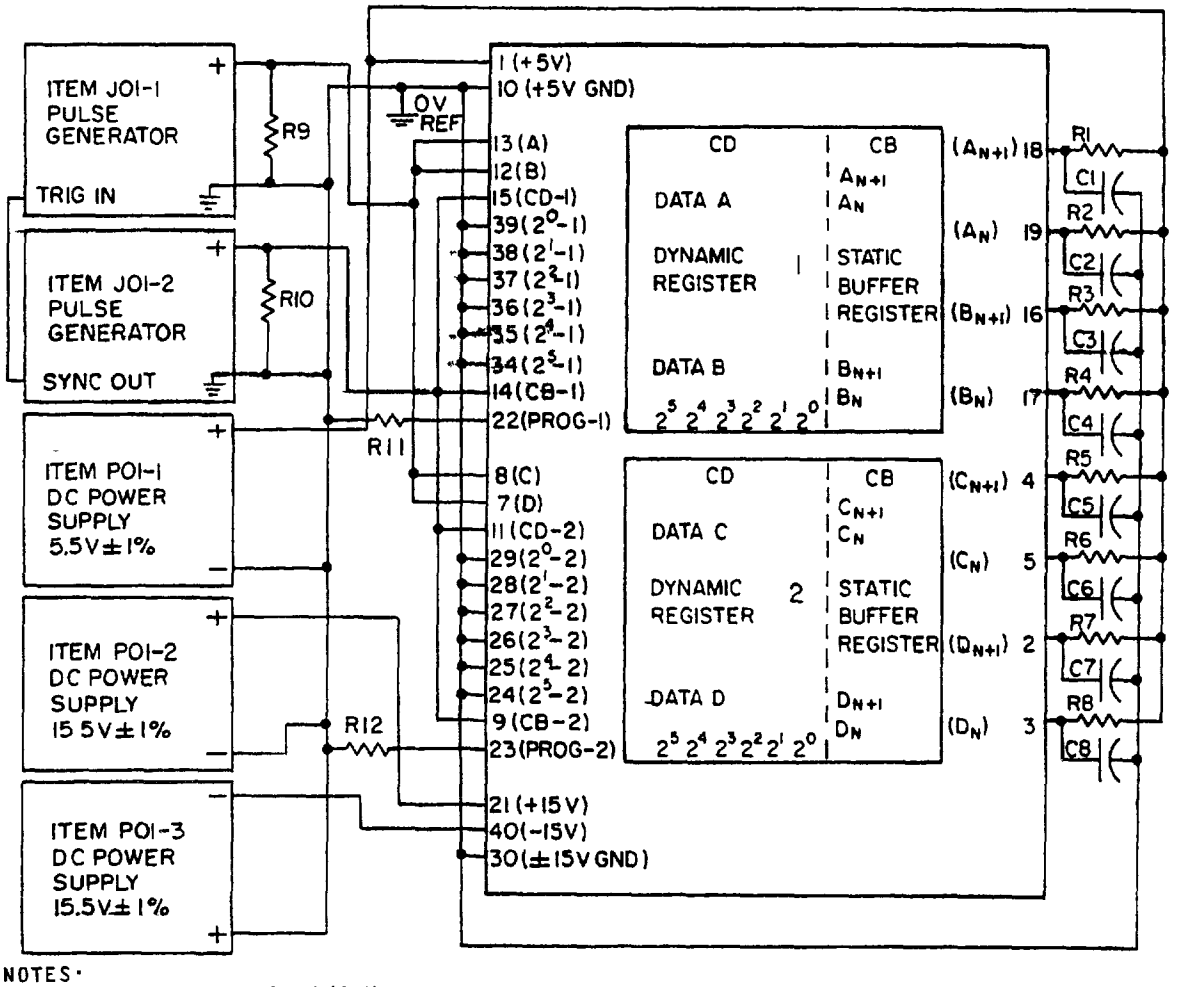
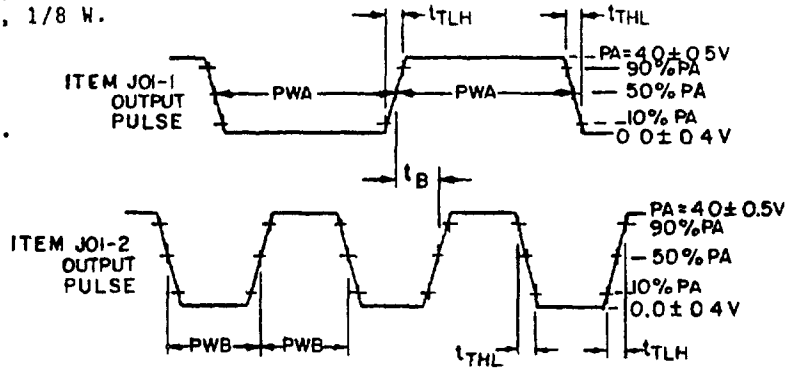


*FIGURE 3 Operating temperature and life test circuit.

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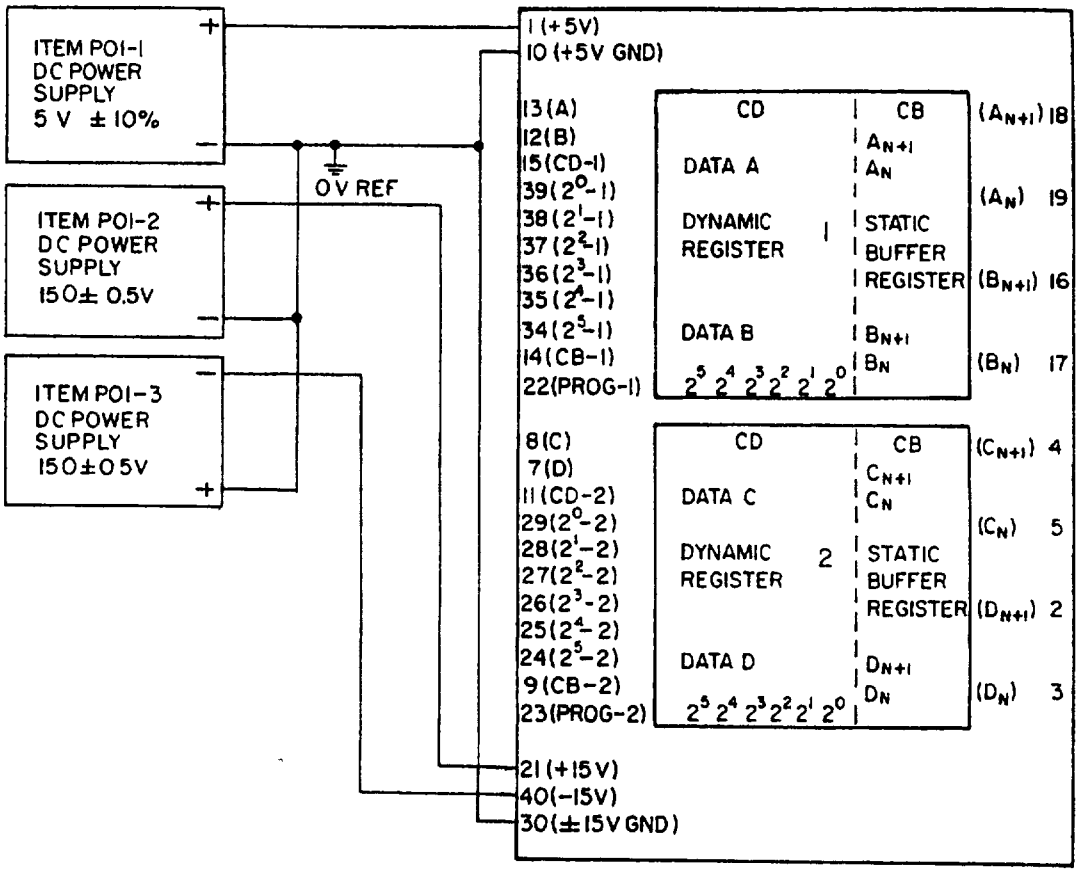


NOTES:
1. R1-R8 = 324Ω ±1%, 1/4 W.
2. R9 = R10 = 51Ω ±5%, 1 W.
3. R11 = R12 = 2.37 kΩ ±1%, 1/8 W.
4. C1-C8 = 100 pF ±20%.
5. Input pulse parameters:
   PWA = 2.0 ±0.2 µs.
   PWB = 1.0 ±0.1 µs.
   t_{TLH} = t_{THL} = 50 ±25 ns.
   t_B = 500 ±100 ns.
6. Refer to handling considerations in 6.6.



* FIGURE 4. Component temperature test circuit.

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NOTE. Refer to handling considerations in 6.6.

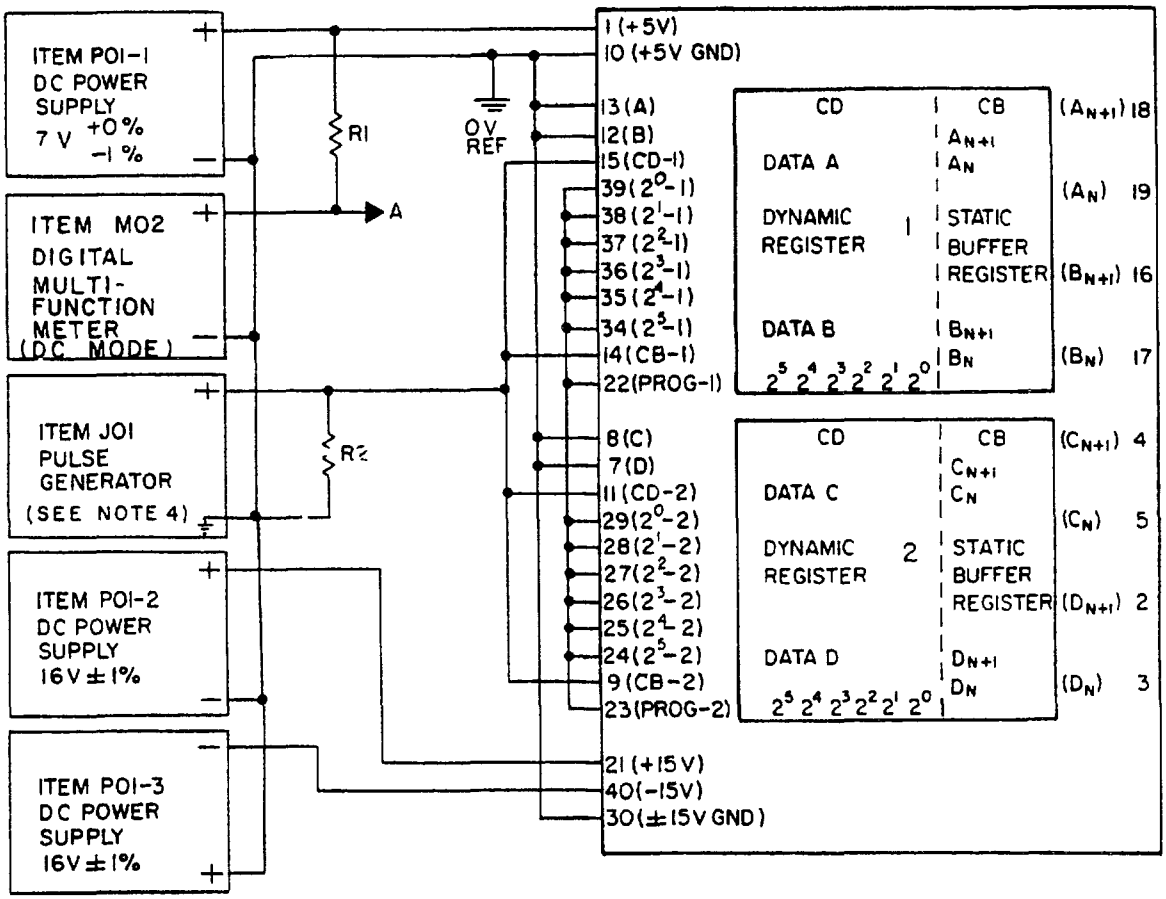
FIGURE 5. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
A1	A_{N+1}	V_{OUT} (low loaded)	18	0	800	mV	6
A2	A_N	"	19	"	"	"	"
A3	B_{N+1}	"	16	"	"	"	"
A4	B_N	"	17	"	"	"	"
A5	C_{N+1}	"	4	"	"	"	"
A6	C_N	"	5	"	"	"	"
A7	D_{N+1}	"	2	"	"	"	"
A8	D_N	"	3	"	"	"	"

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NOTES:
1. R1 = 324Ω ±1%, 1/4 W.
2. R2 =51Ω ±5%. 1 W.
3. Before connecting point A, allow a minimum of 10 µs then connect point A for a minimum of 5 seconds before making measurements.
4. Input J01 setup:
   PW = 1.0 ±0.1 µs.
   PRP = 2.0 ±0.2 µs.
   PA = 0.0 ±0.2 to 4.0 ±0.4 V.
   $t_{TLH} = t_{THL}$ = 50 ±20 ns.
5. Refer to handling considerations in 6.6.

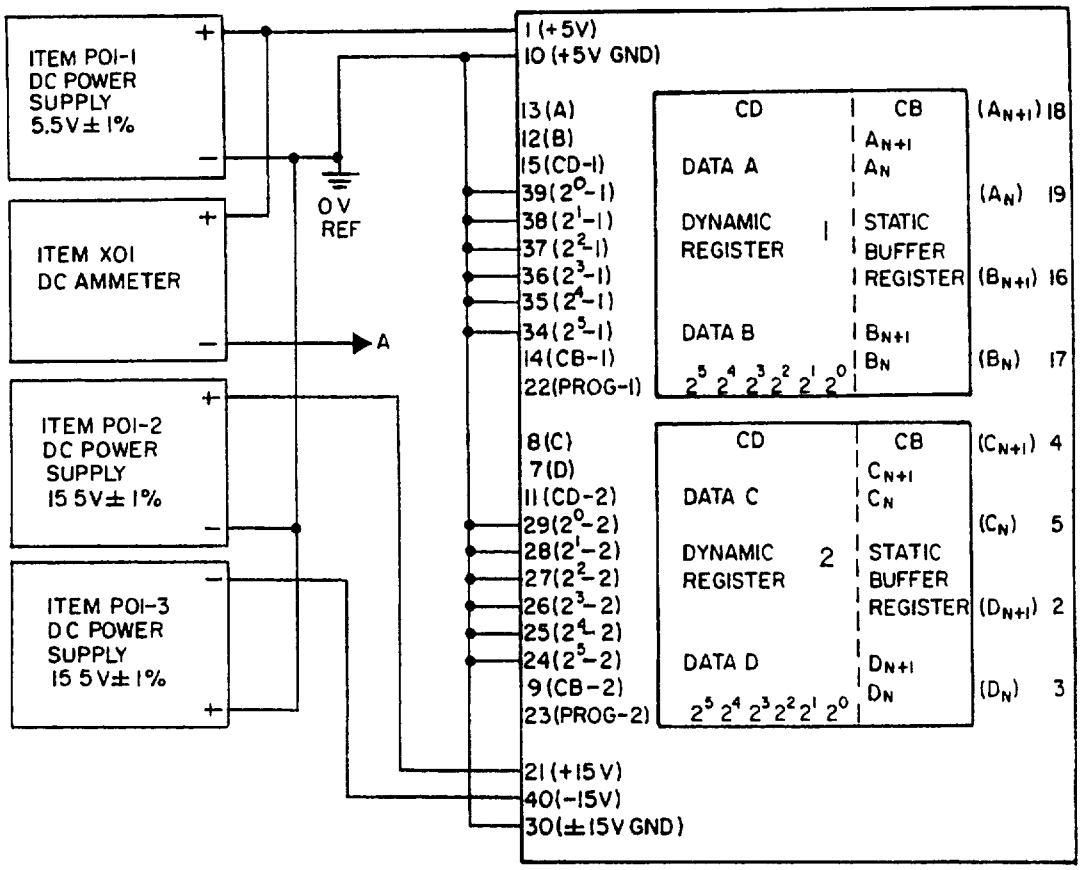
* FIGURE 6. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
B1	A Input breakdown current	13	0	1	mA	7
B2	B "	12	"	"	"	"
B3	C "	8	"	"	"	"
B4	D "	7	"	"	"	"
B5	CD-1 "	15	"	"	"	"
B6	CD-2 "	11	"	"	"	"
B7	CB-1 "	14	"	"	"	"
B8	CB-2 "	9	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: Refer to handling considerations in 6.6.

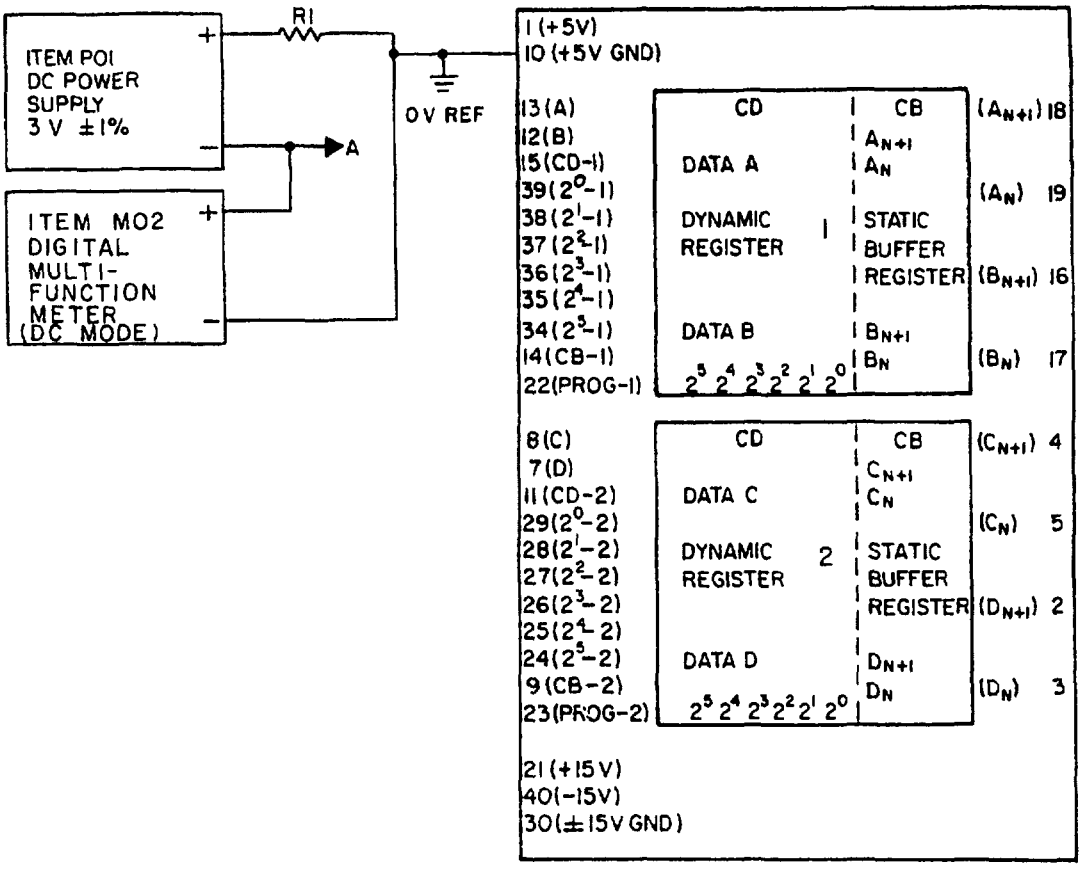
FIGURE 7 Input breakdown current test circuit for table IV

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TABLE V. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
C1	A Input clamp diode voltage	13	-0.4	-1.5	V	8
C2	B "	12	"	"	"	"
C3	C "	8	"	"	"	"
C4	D "	7	"	"	"	"
C5	CD-1 "	15	"	"	"	"
C6	CD-2 "	11	"	"	"	"

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NOTES.
1. R1 = 250Ω ±1%, 1/4 W.
2. Refer to handling considerations in 6.6.

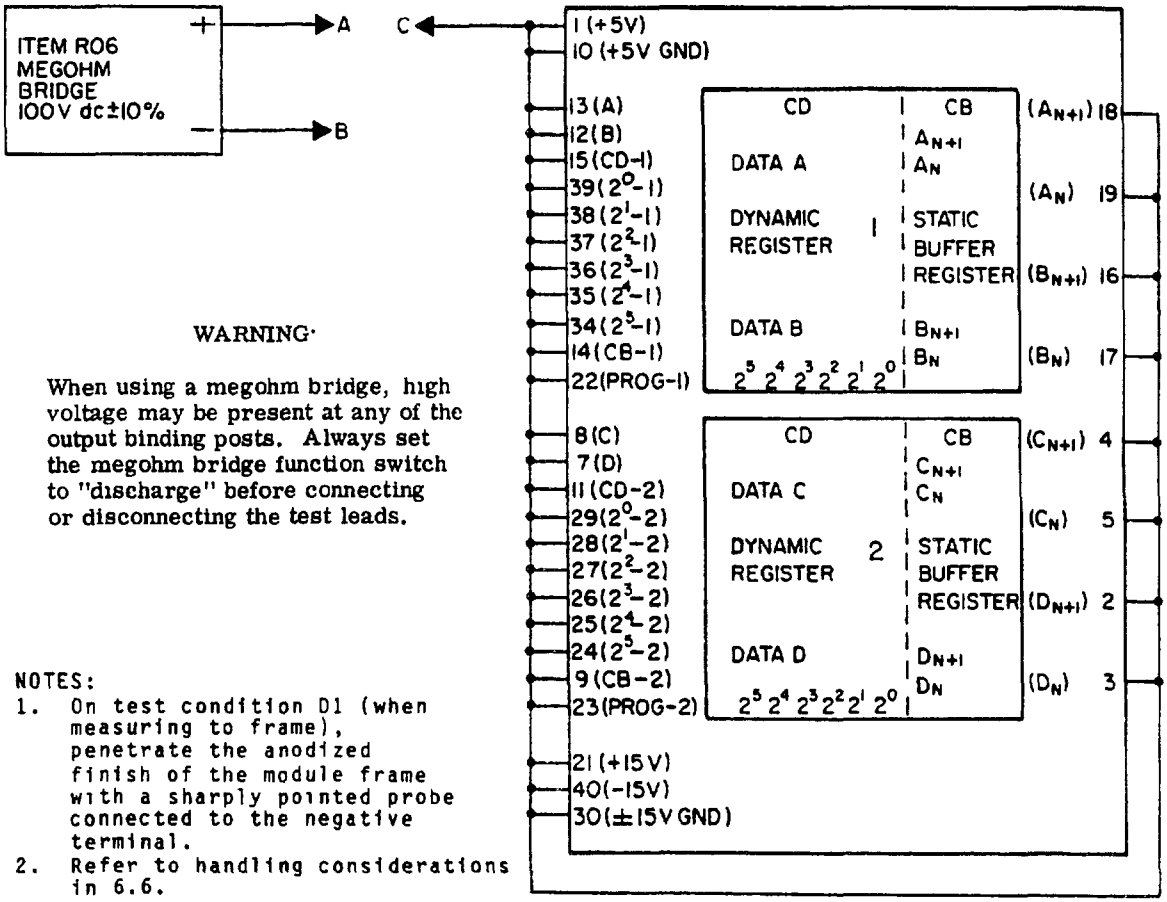
*FIGURE 8. Input clamp diode voltage test circuit for table V

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TABLE VI. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Isolation, frame	C, 6, 20, 33	Frame	10	∞	MΩ	9
D2	Isolation, unused pins	C	6	"	"	"	"
D3	"	"	33	"	"	"	"
D4	"	"	20	"	"	"	"
D5	"	C, 6, 20, 33	Optional pins	"	"	"	"

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTES:
1. On test condition D1 (when measuring to frame), penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.
2. Refer to handling considerations in 6.6.

FIGURE 9 Isolation test circuit for table VI.

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TABLE VII. Output voltage (low-level and high-level) test.

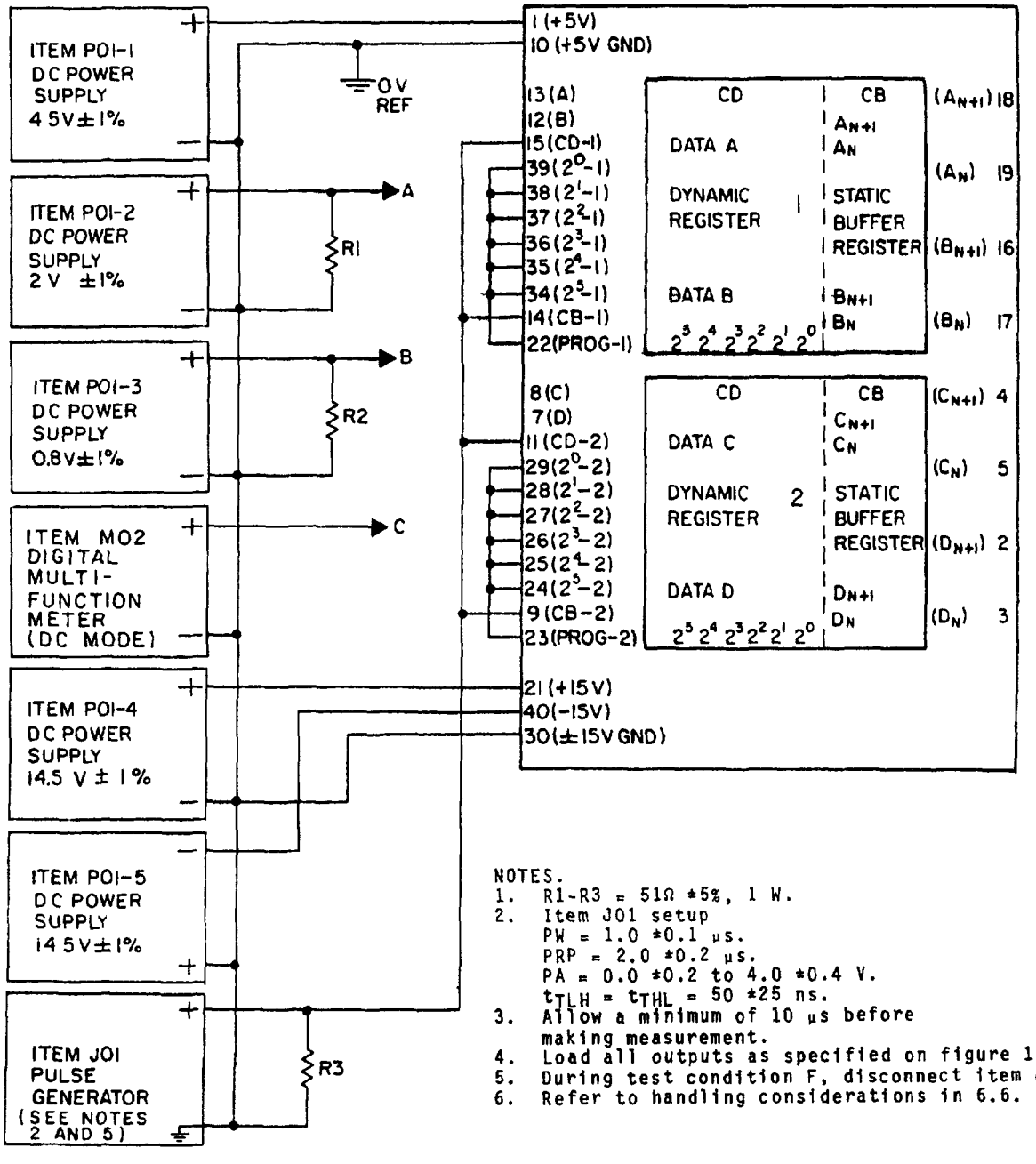
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	A_{N+1}	Output voltage, low level	Open	13,12,8,7	18	0	400	mV	10
E2	A_N	"	"	"	19	"	"	"	"
E3	B_{N+1}	"	"	"	16	"	"	"	"
E4	B_N	"			17	"	"	"	"
E5	C_{N+1}	"	"	"	4	"	"	"	"
E6	C_N	"	"	"	5	"	"	"	"
E7	D_{N+1}	"	"	"	2	"	"	"	"
E8	D_N	"	"	"	3	"	"	"	"
E9	A_{N+1}	Output voltage, high level	13,12,8,7	Open	18	2.4	4.5	V	"
E10	A_N	"	"	"	19	"	"	"	"
E11	B_{N+1}	"	"	"	16	"	"	"	"
E12	B_N	"	"	"	17	"	"	"	"
E13	C_{N+1}	"	"	"	4	"	"	"	"
E14	C_N	"	"	"	5	"	"	"	"
E15	D_N	"	"	"	3	"	"	"	"
E16	D_{N+1}	"	"	"	2	"	"	"	"

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*TABLE VIII. Program output voltage test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F	Program output voltage	Open	Open	22	9.5	14.5	V	10

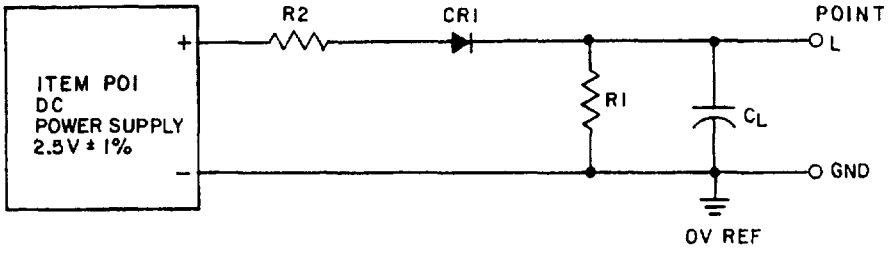
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NOTES.
1. R1-R3 = 51Ω ±5%, 1 W.
2. Item J01 setup
   PW = 1.0 ±0.1 µs.
   PRP = 2.0 ±0.2 µs.
   PA = 0.0 ±0.2 to 4.0 ±0.4 V.
   $t_{TLH} = t_{THL}$ = 50 ±25 ns.
3. Allow a minimum of 10 µs before making measurement.
4. Load all outputs as specified on figure 11.
5. During test condition F, disconnect item J01.
6. Refer to handling considerations in 6.6.

* FIGURE 10. Output voltage (low-level and high-level) and program output voltage test circuit for tables VII and VIII.

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NOTES
1. CR1 = 1N916, or equivalent
2. R1 = 3.01 kΩ ±1%, 1/8 W with a current flow of 800 µA into point L when point L is held at 2.4 Vdc
3. R2 = Select value for current flow of 16 mA ±5% out of point L when point L is held at 0.4 V.
4. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions E, F, L, M, N, P and R.
   = 50 ±5 pF (including probe and parasitic capacitance) for test condition S.
5. C_L is not required for dc electrical tests.
6. The maximum lead length from point L to the appropriate module pin shall be 1 inch (25.4 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 11 Logic load circuit.

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TABLE IX. Input current (low-level and high-level) test.

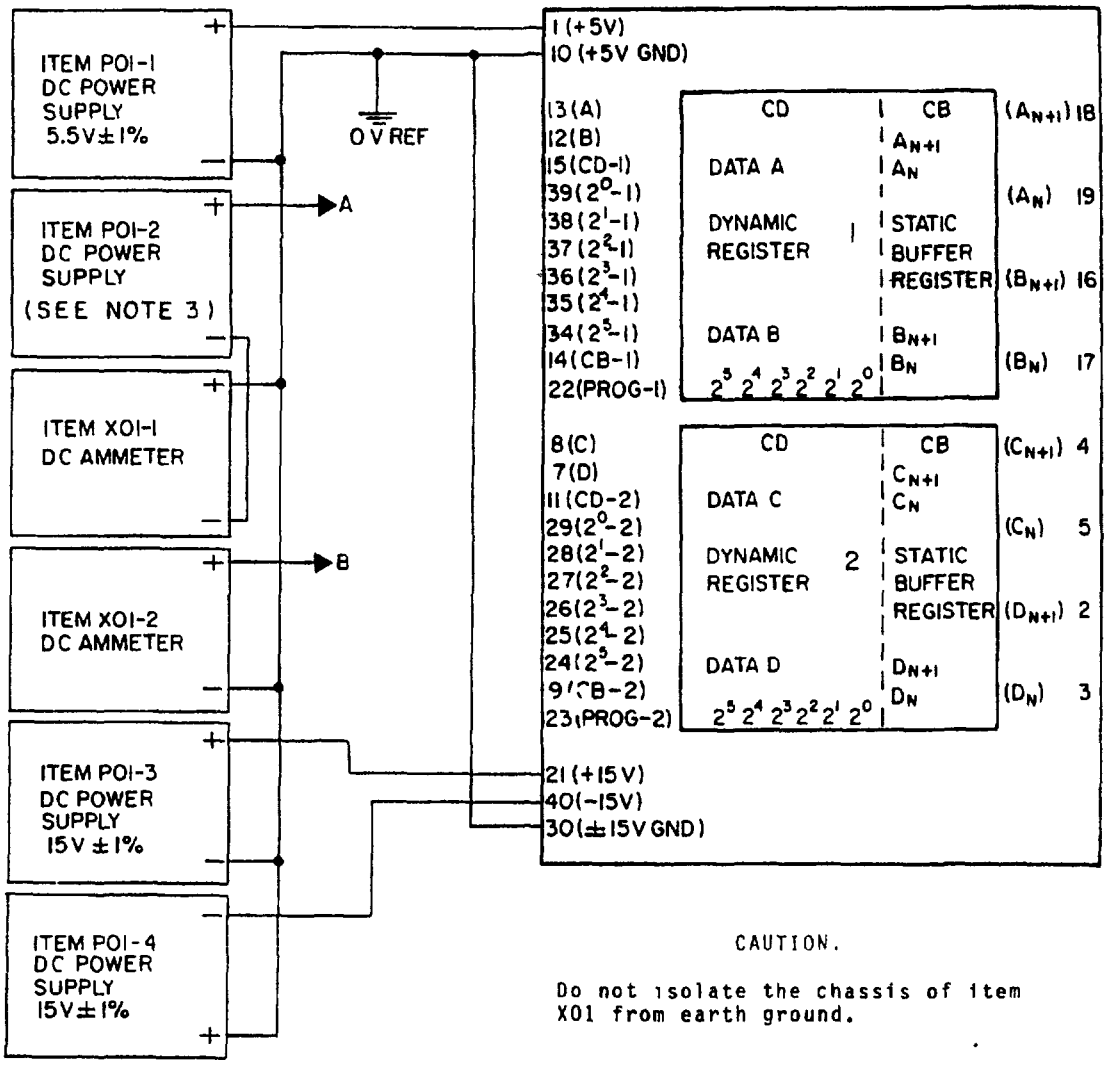
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
G1	A	Input current, low level	Open	13	0.10	1.73	mA	12
G2	B	"	"	12	"	"	"	"
G3	C	"	"	8	"	"	"	"
G4	D	"	"	7	"	"	"	"
G5	CD-1	"	"	15	"	"	"	"
G6	CD-2	"	"	11	"	"	"	"
G7	CB-1	"	"	14	"	"	"	"
G8	CB-2	"	"	9	"	"	"	"
G9	2^0-1	Input current, high level	"	39	0	-100	µA	"
G10	2^1-1	"	"	38	"	"	"	"
G11	2^2-1	"	"	37	"	"	"	"
G12	2^3-1	"	"	36	"	"	"	"
G13	2^4-1	"	"	35	"	"	"	"
G14	2^5-1	"	"	34	"	"	"	"
G15	2^0-2	"	"	29	"	"	"	"
G16	2^1-2	"	"	28	"	"	"	"
G17	2^2-2	"	"	27	"	"	"	"
G18	2^3-2	"	"	26	"	"	"	"
G19	2^4-2	"	"	25	"	"	"	"
G20	2^5-2	"	"	24	"	"	"	"

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*TABLE X. Input current (high-level and low-level) test.

Test condition	Parameter		Connect		Initial limits 25 ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
H1	A	Input current, high level	13	Open	0	40	µA	12
H2	B	"	12	"	"	"	"	"
H3	C	"	8	"	"	"	"	"
H4	D	"	7	"	"	"	"	"
H5	CD-1	"	15	"	"	"	"	"
H6	CD-2	"	11	"	"	"	"	"
H7	CB-1	"	14	"	"	"	"	"
H8	CB-2	"	9	"	"	"	"	"
H9	2^0-1	Input current, low level	39	"	"	300	"	"
H10	2^1-1	"	38	"	"	"	"	"
H11	2^2-1	"	37	"	"	"	"	"
H12	2^3-1	"	36	"	"	"	"	"
H13	2^4-1	"	35	"	"	"	"	"
H14	2^5-1	"	34	"	"	"	"	"
H15	2^0-2	"	29	"	"	"	"	"
H16	2^1-2	"	28	"	"	"	"	"
H17	2^2-2	"	27	"	"	"	"	"
H18	2^3-2	"	26	"	"	"	"	"
H19	2^4-2	"	25	"	"	"	"	"
H20	2^5-2	"	24	"	"	"	"	"

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NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. With point A disconnected, item X01-1 should indicate zero current on the 1 µA range.
3. Adjust item P01-2 to 2.4 V ±1% for test conditions H1-H8 and 12 V ±1% for test conditions H9-H20.
4. Refer to handling considerations in 6.6.

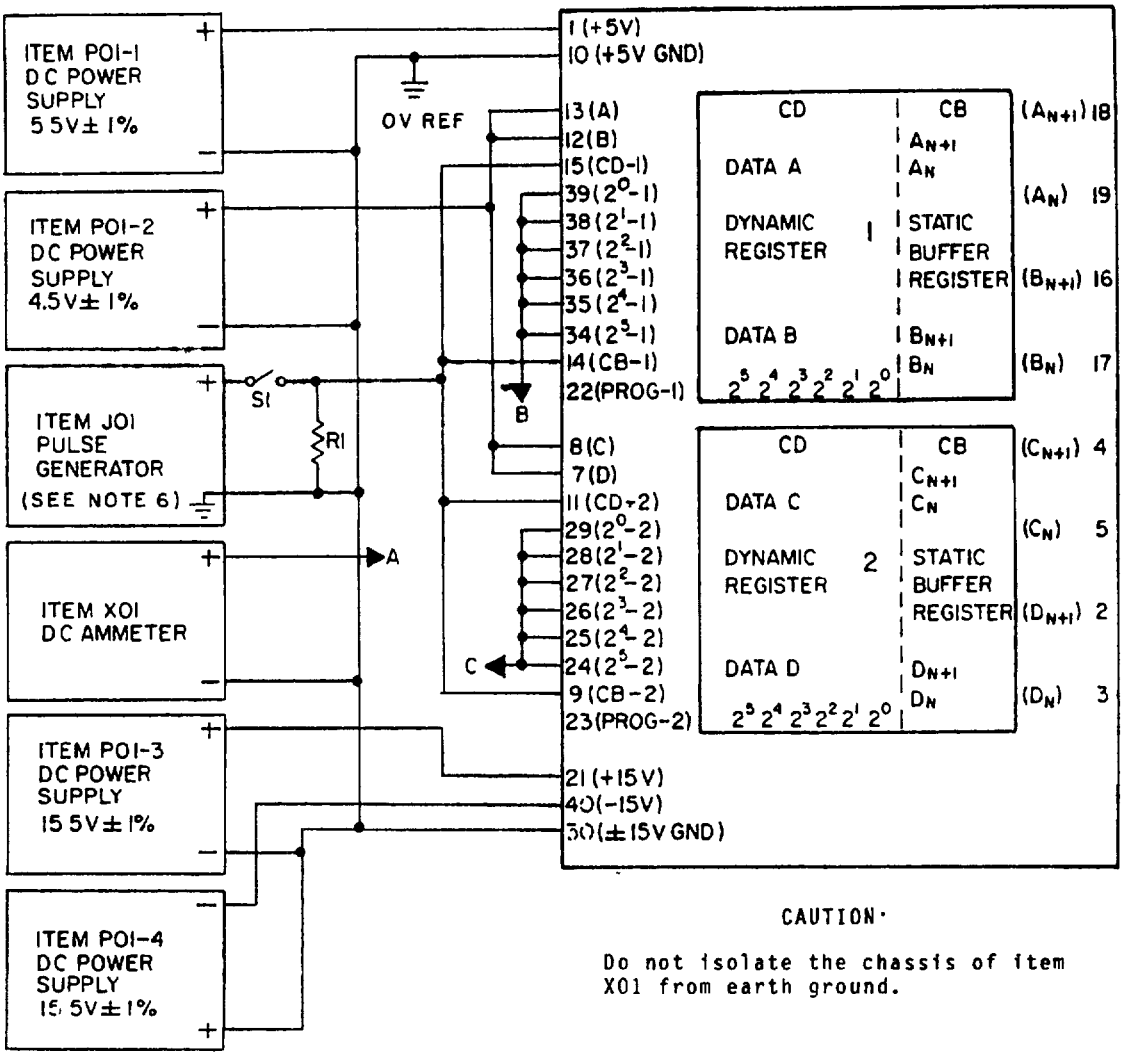
* FIGURE 12 Input current (low-level and high-level) test circuit for tables IX and X

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TABLE XI. Output short circuit current test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J1	A_{N+1}	Output short circuit current	18	22	Open	18	57	mA	13
J2	A_N	"	19	"	"	"	"	"	"
J3	B_{N+1}	"	16	"	"	"	"	"	"
J4	B_N	"	17	"	"	"	"	"	"
J5	C_{N+1}	"	4	Open	23	"	"	"	"
J6	C_N	"	5	"	"	"	"	"	"
J7	D_{N+1}	"	2	"	"	"	"	"	"
J8	D_N	"	3	"	"	"	"	"	"
J9	PROG	Output short circuit current, program	22	"	Open	19	24	"	"

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NOTES
1. R1 = 51Ω ±5%, 1 W.
2. S1 = SPST switch.
3. When performing output short circuit tests only one output shall be shorted at a time and the measurement taken within 5 seconds.
4. Close S1 for a minimum of 10 µs then make measurement.
5. With point A disconnected, item X01 should indicate zero current.
6. Item J01 setup:
   PW = 1.0 ±0.1 µs.
   PRP = 2.0 ±0.2 µs.
   PA = 0.0 ±0.2 to 4.0 ±0.4 V.
   t_{TLH} = t_{THL} = 50 ±25 ns.
7. Refer to handling considerations in 6.6.

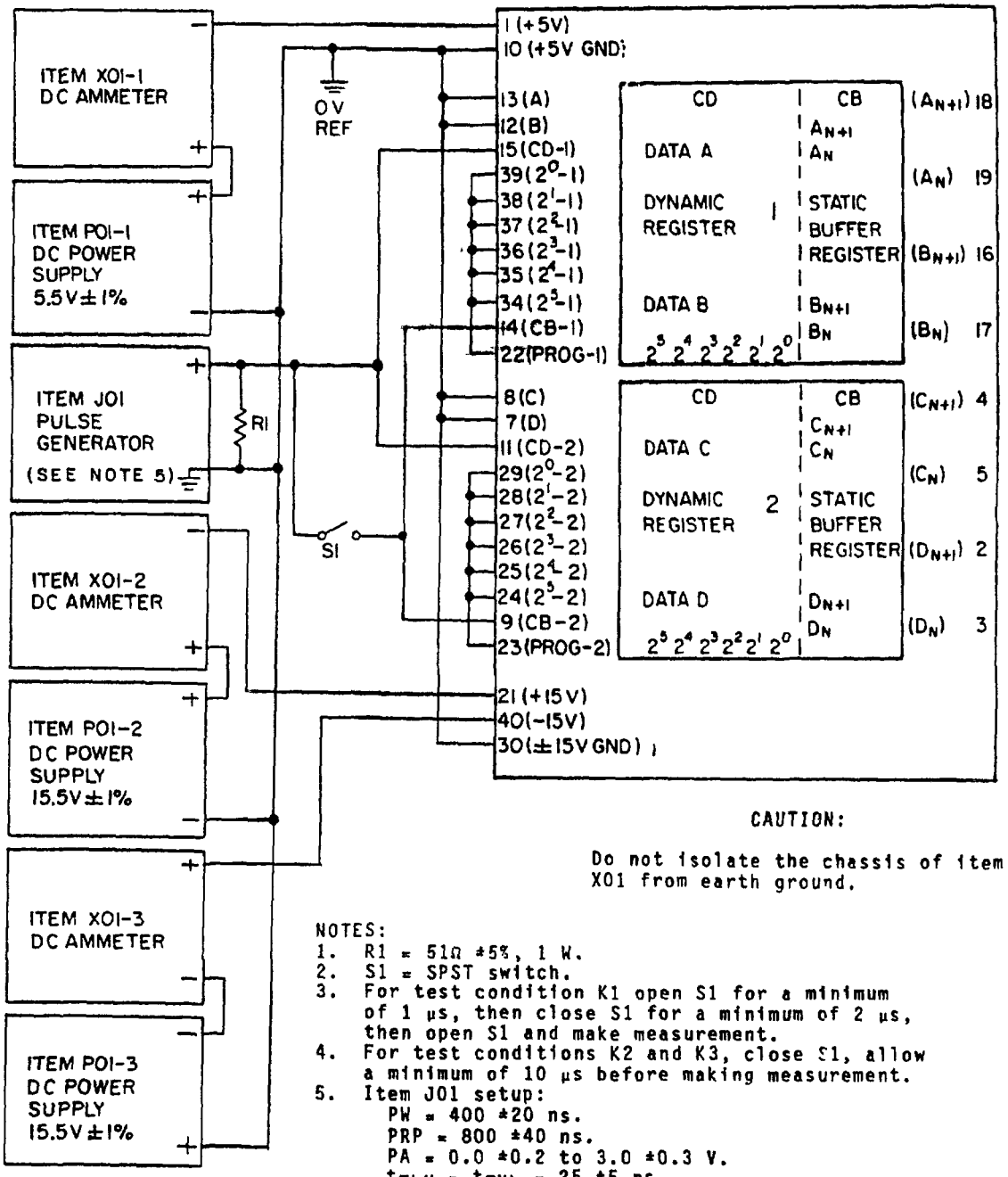
FIGURE 13 Output short circuit current test circuit for table XI.

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*TABLE XII. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
K1	Power supply current (5.5 V)	5	150	mA	14
K2	Power supply current (15.5 V at 1.2 MHz clock rate)	5	100	"	"
K3	Power supply current (-15.5 V at 1.2 MHz clock rate)	0	30	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. S1 = SPST switch.
3. For test condition K1 open S1 for a minimum of 1 µs, then close S1 for a minimum of 2 µs, then open S1 and make measurement.
4. For test conditions K2 and K3, close S1, allow a minimum of 10 µs before making measurement.
5. Item J01 setup:
   PW = 400 ±20 ns.
   PRP = 800 ±40 ns.
   PA = 0.0 ±0.2 to 3.0 ±0.3 V.
   $t_{TLH} = t_{THL} = 25 \pm 5$ ns.
6. Refer to handling considerations in 6.6.

* FIGURE 14. Power supply current test circuit for table XII.

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*TABLE XIII. Maximum CD clock pulse width check and input hold time check test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	Maximum CD clock pulse width check	13	15,14	19	Open	180	220	µs	15,16
L2	"	12	15,14	17	"	"	"	"	"
L3	"	8	11,9	5	"	"	"	"	"
L4	"	7	11,9	3	"	"	"	"	"
L5	Input hold time check	13	15,14	19	"	1560	1640	ns	15,17
L6	"	12	15,14	17	"	"	"	"	"
L7	"	8	11,9	5	"	"	"	"	"
L8	"	7	11,9	3	"	"	"	"	"
L9	"	13	15,14	19	"	"	"	"	"
L10	"	12	15,14	17	"	"	"	"	"
L11	"	8	11,9	5	"	"	"	"	"
L12	"	7	11,9	3	"	"	"	"	"

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TABLE XIV. Transition time (t_{TLH} and T_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	A_{N+1} Transition time (t_{TLH})	13	15,14	18	Open	4	50	ns	15,18
M2	A_N "	13	"	19	"	"	"	"	"
M3	B_{N+1} "	12	"	16	"	"	"	"	"
M4	B_N "	12	"	17	"	"	"	"	"
M5	C_{N+1} "	8	11,9	4	"	"	"	"	"
M6	C_N "	8	"	5	"	"	"	"	"
M7	D_{N+1} "	7	"	2	"	"	"	"	"
M8	D_N "	7	"	3	"	"	"	"	"
M9	A_{N+1} Transition time (t_{THL})	13	15,14	18	"	"	20	"	"
M10	A_N "	13	"	19	"	"	"	"	"
M11	B_{N+1} "	12	"	16	"	"	"	"	"
M12	B_N "	12	15,14	17	"	"	"	"	"
M13	C_{N+1} "	8	11,9	4	"	"	"	"	"
M14	C_N "	8	"	5	"	"	"	"	"
M15	D_{N+1} "	7	"	2	"	"	"	"	"
M16	D_N "	7	"	3	"	"	"	"	"

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*TABLE XV. Propagation delay time (t_{PLH}) test.

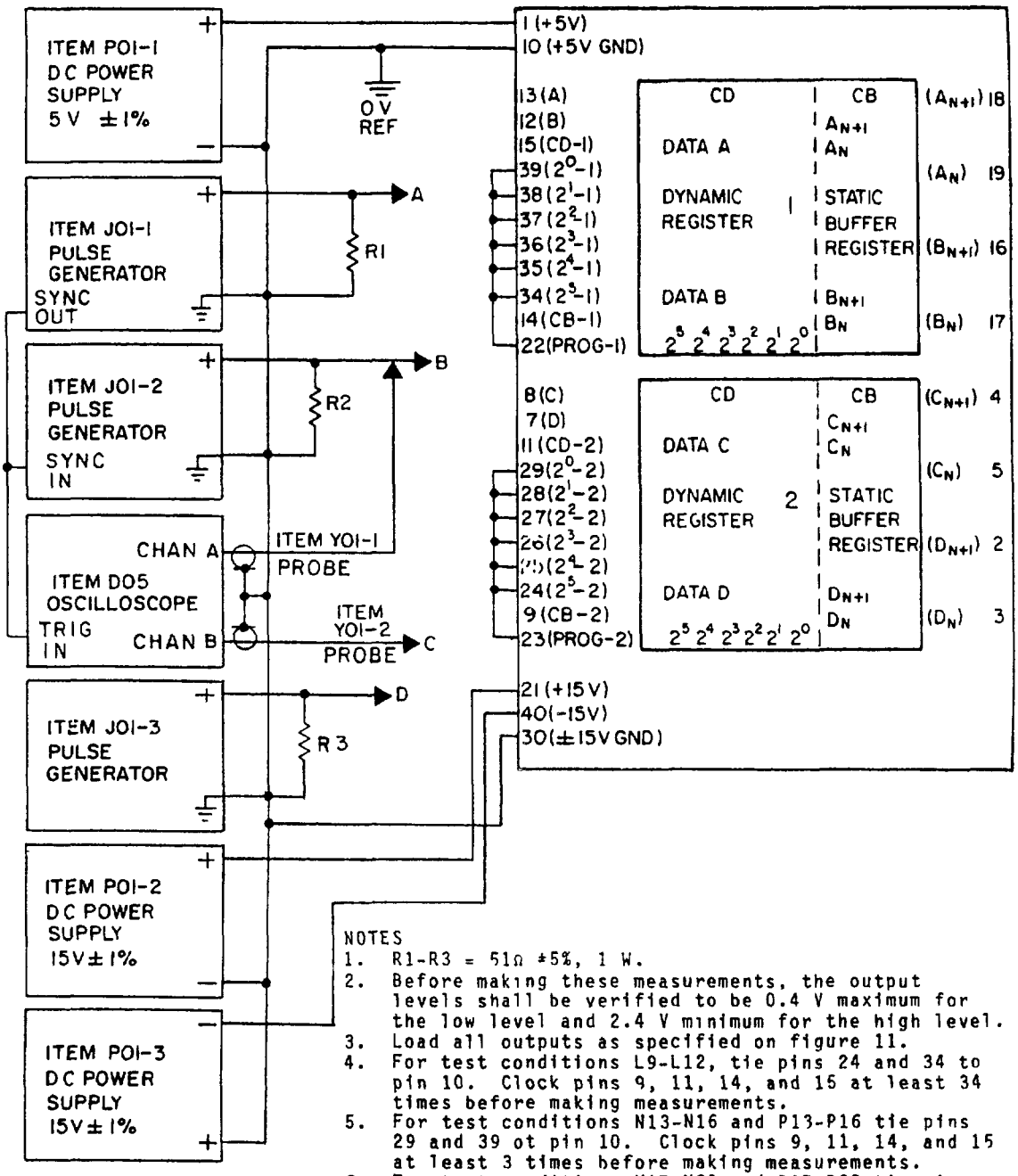
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	Propagation delay time, t_{PLH} (CB clock to output)	13	15,14	18	Open	10	45	ns	15,18
N2	"	13	"	19	"	"	"	"	"
N3	"	12	"	16	"	"	"	"	"
N4	"	12	"	17	"	"	"	"	"
N5	"	8	11,9	4	"	"	"	"	"
N6	"	8	"	5	"	"	"	"	"
N7	"	7	"	2	"	"	"	"	"
N8	"	7	"	3	"	"	"	"	"
N9	Propagation delay time, t_{PLH} (CD clock to output)	13	15	19	14	50	400	"	15,19
N10	"	12	15	17	14	"	"	"	"
N11	"	8	11	5	9	"	"	"	"
N12	"	7	11	3	9	"	"	"	"
N13	Propagation delay time, t_{PLH} (CB clock to output)	13	15,14	19	Open	10	45	"	15,18
N14	"	12	15,14	17	"	"	"	"	"
N15	"	8	11,9	5	"	"	"	"	"
N16	"	7	11,9	3	"	"	"	"	"
N17	"	13	15,14	19	"	"	"	"	"
N18	"	12	15,14	17	"	"	"	"	"
N19	"	8	11,9	5	"	"	"	"	"
N20	"	7	11,9	3	"	"	"	"	"

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*TABLE XVI. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	Propagation delay time, t_{PHL} (CB clock to output)	13	15,14	18	Open	10	45	ns	15,18
P2	"	13	"	19	"	"	"	"	"
P3	"	12	"	16	"	"	"	"	"
P4	"	12	"	17	"	"	"	"	"
P5	"	8	11,9	4	"	"	"	"	"
P6	"	8	"	5	"	"	"	"	"
P7	"	7	"	2	"	"	"	"	"
P8	"	7	"	3	"	"	"	"	"
P9	Propagation delay time, t_{PHL} (CD clock to output)	13	15	19	14	50	400	"	15,19
P10	"	12	15	17	14	"	"	"	"
P11	"	8	11	5	9	"	"	"	"
P12	"	7	11	3	9	"	"	"	"
P13	Propagation delay time, t_{PHL} (CB clock to output)	13	15,14	19	Open	10	45	"	15,18
P14	"	12	15,14	17	"	"	"	"	"
P15	"	8	11,9	5	"	"	"	"	"
P16	"	7	11,9	3	"	"	"	"	"
P17	"	13	15,14	19	"	"	"	"	"
P18	"	12	15,14	17	"	"	"	"	"
P19	"	8	11,9	5	"	"	"	"	"
P20	"	7	11,9	3	"	"	"	"	"

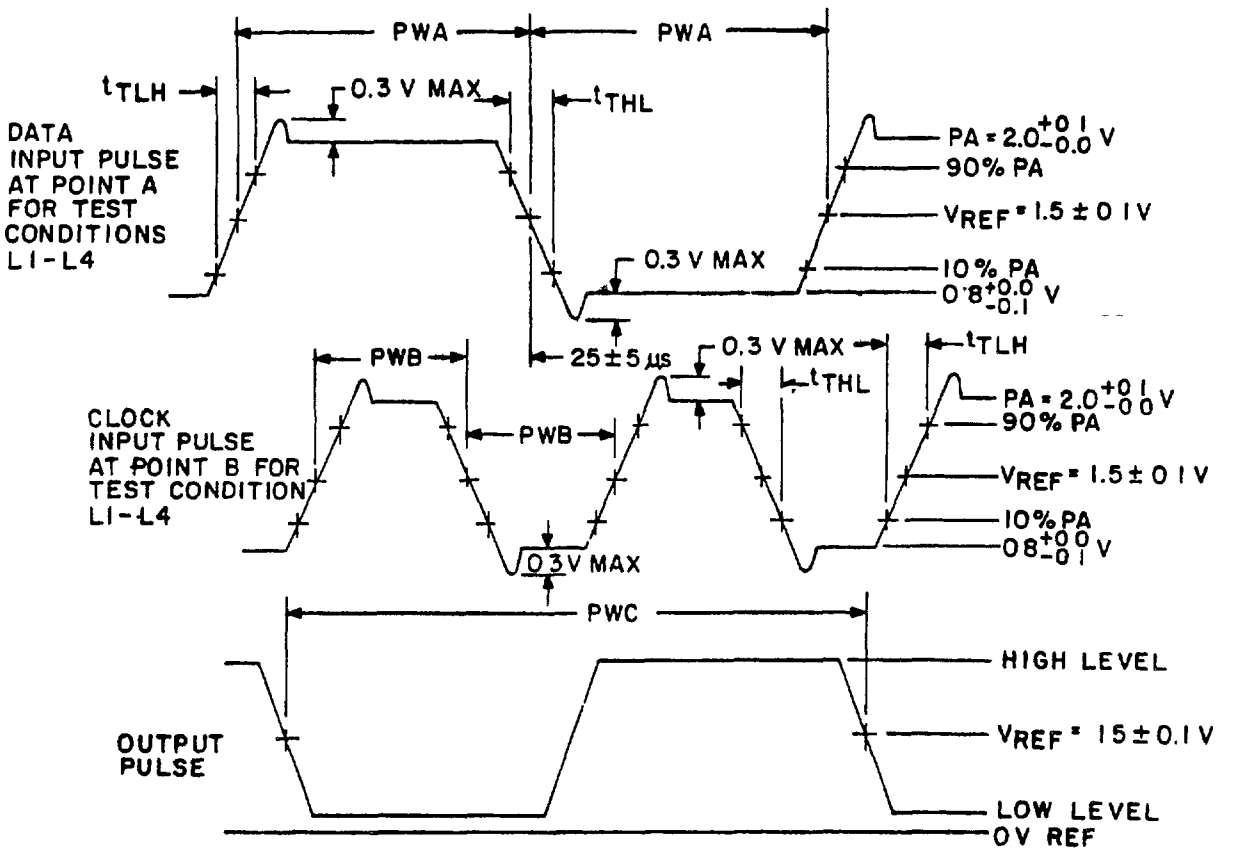
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NOTES
1. R1-R3 = 51Ω ±5%, 1 W.
2. Before making these measurements, the output levels shall be verified to be 0.4 V maximum for the low level and 2.4 V minimum for the high level.
3. Load all outputs as specified on figure 11.
4. For test conditions L9-L12, tie pins 24 and 34 to pin 10. Clock pins 9, 11, 14, and 15 at least 34 times before making measurements.
5. For test conditions N13-N16 and P13-P16 tie pins 29 and 39 ot pin 10. Clock pins 9, 11, 14, and 15 at least 3 times before making measurements.
6. For test conditions N17-N20 and P17-P20 tie pins 28 and 38 to pin 10. Clock pins 9, 11, 14, and 15 at least 4 times before making measurements.
7. Refer to handling consideration in 6.6.

* FIGURE 15. Maximum CD clock pulse width check and input hold time check, transition time (t_{TLH} and t_{THL}), and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XIII, XIV, XV, and XVI.

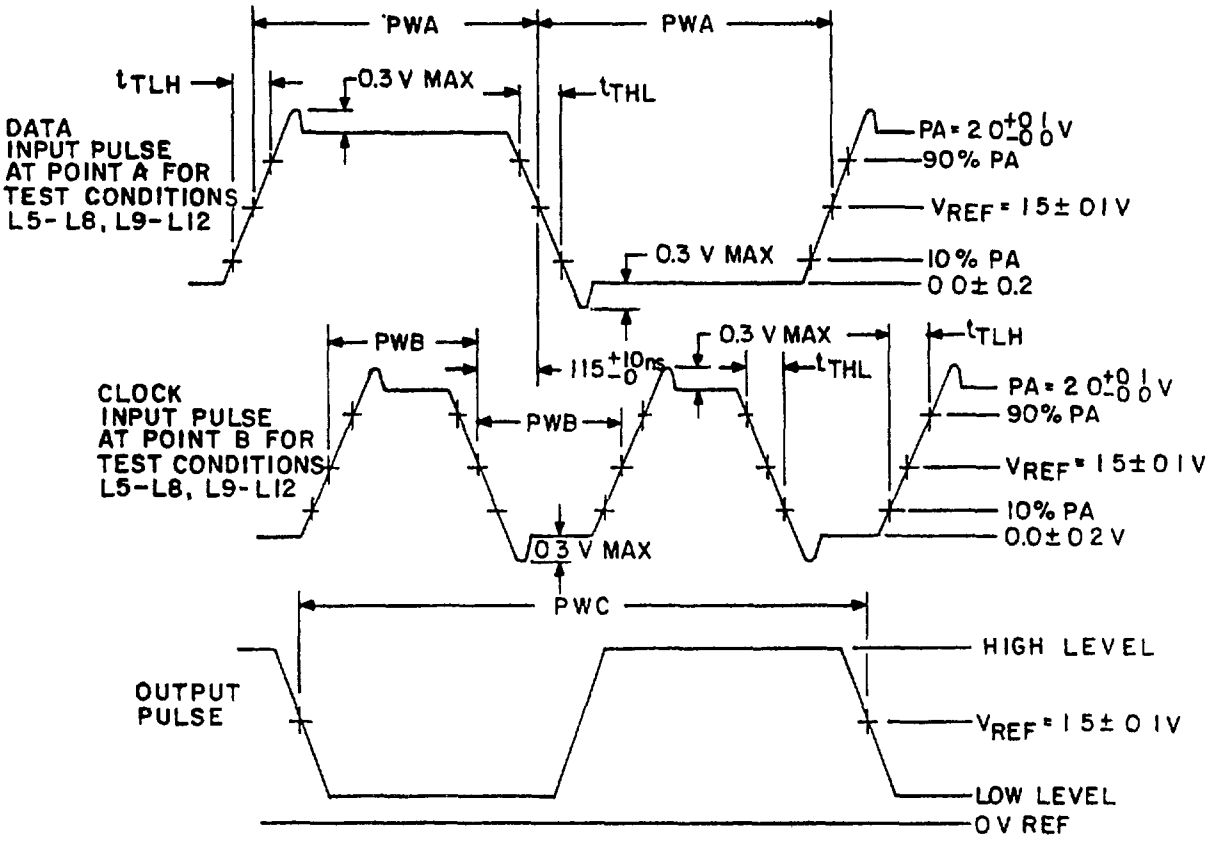
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NOTES:
1. Point A input pulse parameters:
   PWA = 100 ±1 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Point B input pulse parameters:
   PWB = 50.0 ±0.5 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
3. PWC = as specified in table XIII.

* FIGURE 16. Maximum CD clock pulse width check test circuit waveforms for table XIII.

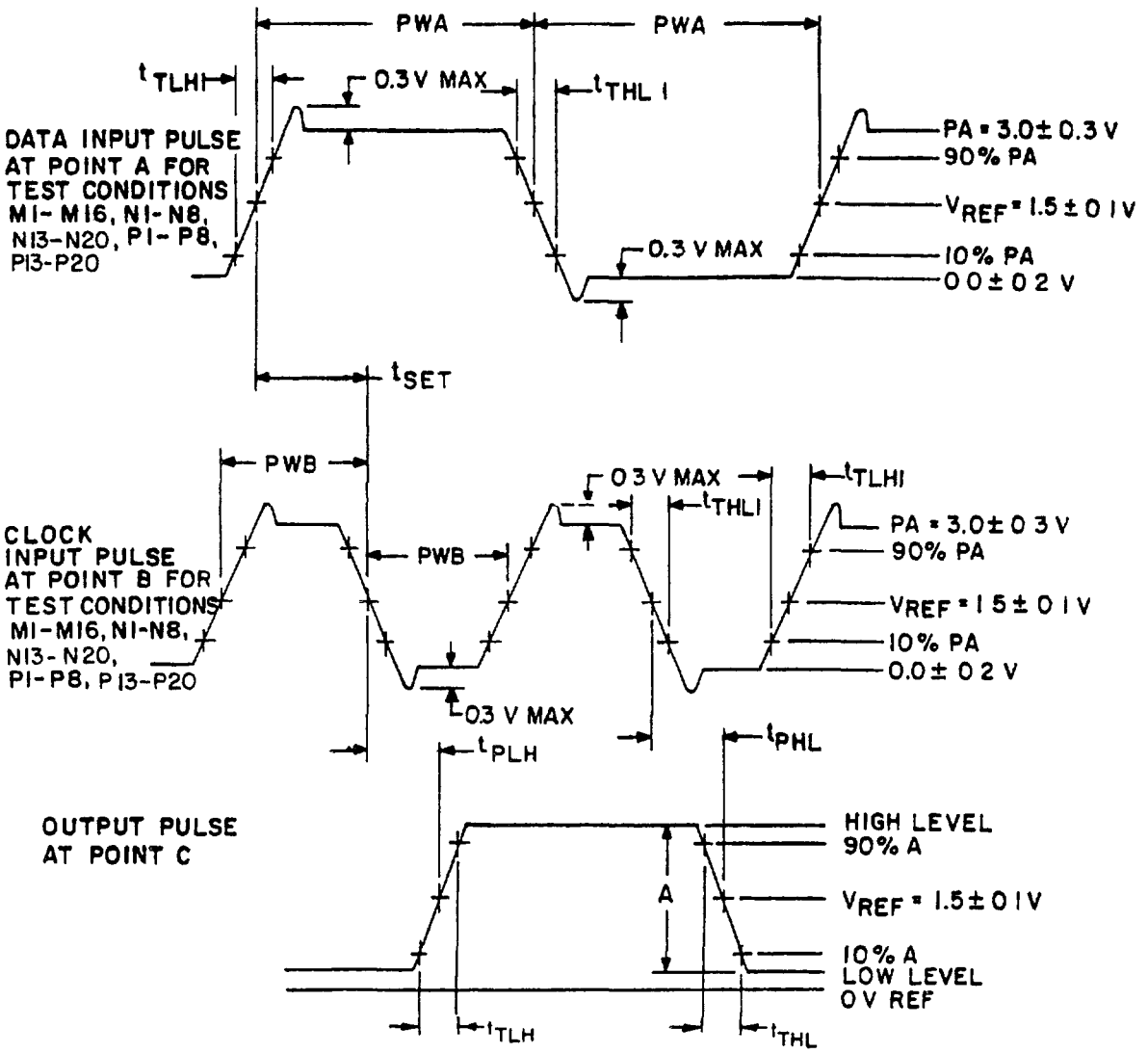
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NOTES
1. Point A input pulse parameters
   PWA = 800 +20 -0 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Point B input pulse parameters
   PWB = 400 +20 -0 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
3. This test verifies input hold time.

* FIGURE 17. Input hold time check test circuit waveforms for table XIII.

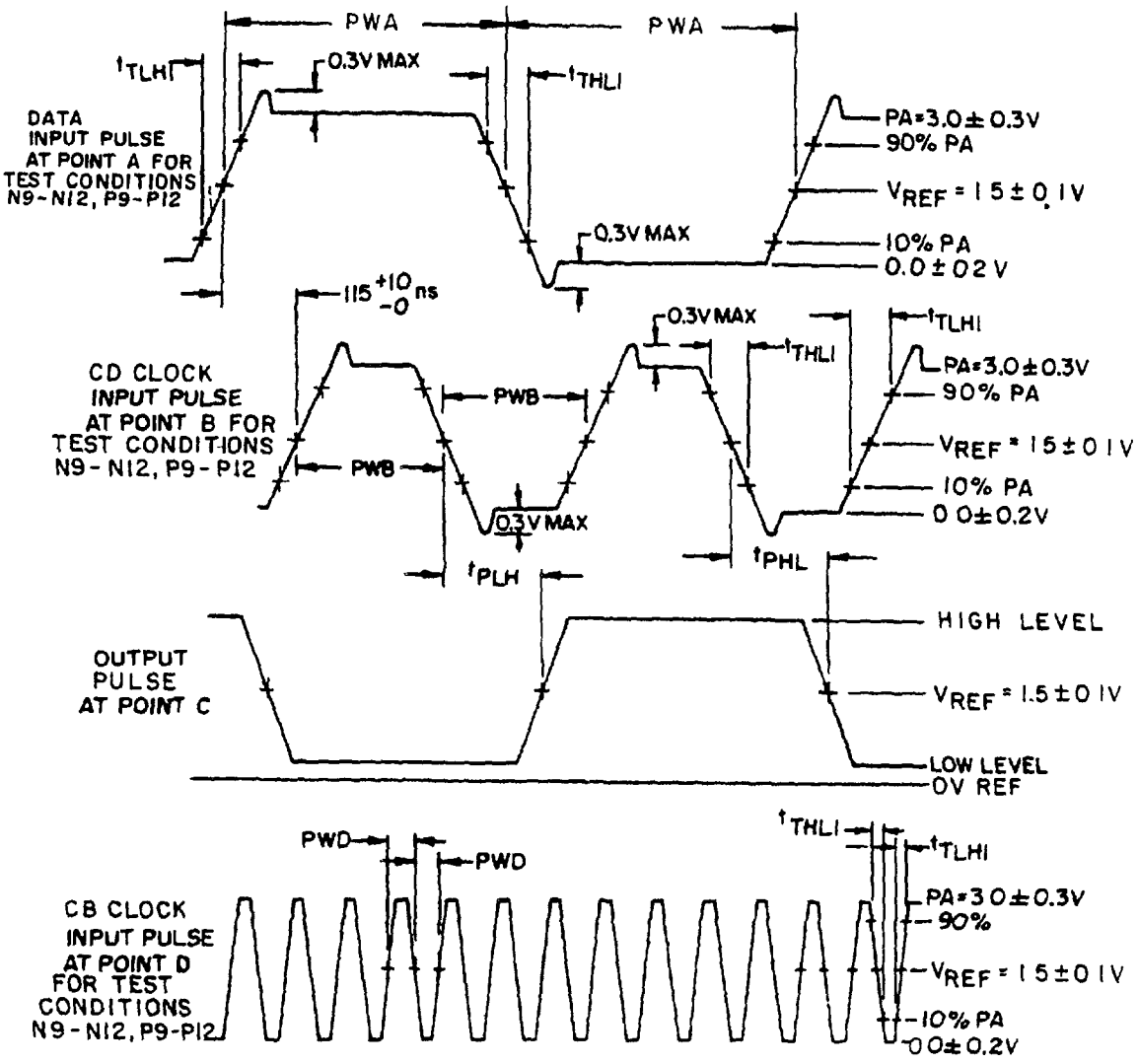
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NOTES:
1. Point A input pulse parameters:
   PWA = 800 +20, -0 ns.
   t_{TLH1} = t_{THL1} = 10 ±2 ns.
2. Point B input pulse parameters:
   PWB = 400 +20, -0 ns.
   t_{TLH1} = t_{THL1} = 10 ±2 ns.
3. These tests also verify minimum data setup time, maximum CD clock frequency, and simultaneous clocking.
4. t_{SET} = 115 +10, -0 ns for test conditions N17-N20 and P17-P20.
   = 200 +10, -0 ns for test conditions M1-M16, N1-N8 and P1-P8.
   = 135 +10, -0 ns for test conditions N13-N16 and P13-P16.
5. All voltage levels are referenced to ground.

* FIGURE 18. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIV, XV, and XVI.

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NOTES:
1. Point A input pulse parameters
   PWA = 800 +20, -0 ns.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
2. Point B input pulse parameters:
   PWB = 400 +20, -0 ns.
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
3. Point D input pulse parameters
   PWD = 20 +2, -0 ns.
   $t_{TLH1} = t_{THL1} = 6 \pm 2$ ns.
4. These tests also verify minimum setup times and maximum CD and CB clock frequency.
5. All voltage levels are referenced to ground.

* FIGURE 19. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XV and XVI.

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TABLE XVII. Dynamic shift register operation.

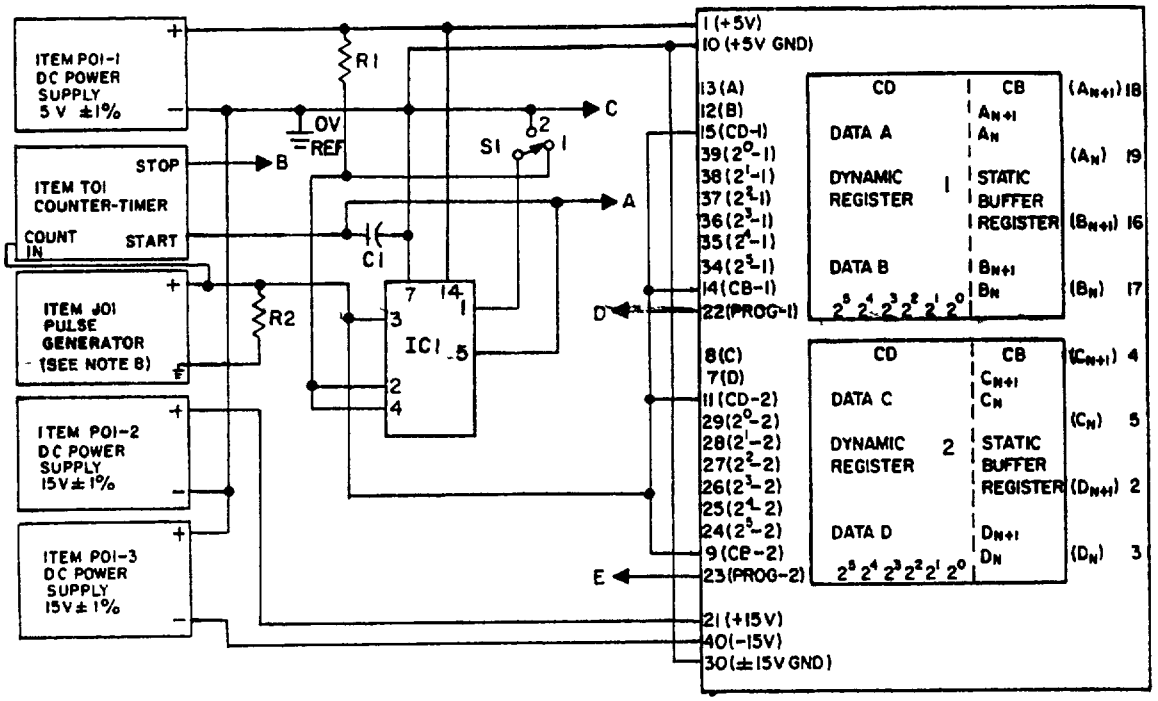
Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R1	1 BIT	Register operation	13	19	Open	34,35,36,37,38,39	Open	2	2	Count	20
R2	1 BIT	"	12	17	"	34,35,36,37,38,39	"	"	"	"	"
R3	1 BIT	"	8	5	"	Open	24,25,26,27,28,29	"	"	"	"
R4	1 BIT	"	7	3	"	Open	24,25,26,27,28,29	"	"	"	"
R5	2 BIT	"	13	19	39	34,35,36,37,38	Open	3	3	"	"
R6	2 BIT	"	12	17	39	34,35,36,37,38	Open	"	"	"	"
R7	2 BIT	"	8	5	29	Open	24,25,26,27,28	"	"	"	"
R8	2 BIT	"	7	3	29	Open	24,25,26,27,28	"	"	"	"
R9	3 BIT	"	13	19	38	34,35,36,37,39	Open	4	4	"	"
R10	3 BIT	"	12	17	38	34,35,36,37,39	Open	"	"	"	"
R11	3 BIT	"	8	5	28	Open	24,25,26,27,29	"	"	"	"
R12	3 BIT	"	7	3	28	Open	24,25,26,27,29	"	"	"	"
R13	5 BIT	"	13	19	37	34,35,36,38,39	Open	6	6	"	"
R14	5 BIT	"	12	17	37	34,35,36,38,39	Open	"	"	"	"
R15	5 BIT	"	8	5	27	Open	24,25,26,28,29	"	"	"	"
R16	5 BIT	"	7	3	27	Open	24,25,26,28,29	"	"	"	"
R17	9 BIT	"	13	19	36	34,35,37,38,39	Open	10	10	"	"
R18	9 BIT	"	12	17	36	34,35,37,38,39	Open	"	"	"	"
R19	9 BIT	"	8	5	26	Open	24,25,27,28,29	"	"	"	"
R20	9 BIT	"	7	3	26	Open	24,25,27,28,29	"	"	"	"

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TABLE XVII. Dynamic shift register operation - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R21	17 BIT Register operation	13	19	35	34,36,37,38,39	Open	18	18	Count	20
R22	17 BIT "	12	17	35	34,36,37,38,39	Open	"	"	"	"
R23	17 BIT "	8	5	25	Open	24,26,27,28,29	"	"	"	"
R24	17 BIT "	7	3	25	Open	24,26,27,28,29	"	"	"	"
R25	33 BIT "	13	19	34	35,36,37,38,39	Open	34	34	"	"
R26	33 BIT "	12	17	34	35,36,37,38,39	Open	"	"	"	"
R27	33 BIT "	8	5	24	Open	25,26,27,28,29	"	"	"	"
R28	33 BIT "	7	3	24	"	25,26,27,28,29	"	"	"	"
R29	64 BIT "	13	19	34,35,36,37,38,39	"	Open	65	65	"	"
R30	64 BIT "	12	17	34,35,36,37,38,39	"	"	"	"	"	"
R31	64 BIT "	8	5	24,25,26,27,28,29	"	"	"	"	"	"
R32	64 BIT "	7	3	24,25,26,27,28,29	"	"	"	"	"	"

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NOTES:
1. R1 = 1 kΩ ±1%, 1/8 W.
2. R2 = 51Ω ±5%, 1 W.
3. S1 = SPDT switch.
4. IC1 = 7474 (J, N, W) or equivalent.
5. C1 = 100 pF ±20%.
6. Adjust item T01 for start and stop on positive edge of pulse.
7. For each test condition.
   a. S1 to be in position 2.
   b. Reset counter.
   c. After minimum of 70 µs, place S1 in position 1.
   d. When totalizing is complete, read measurement of item T01.
8. Item J01 setup:
   PW = 500 ±20 ns.
   PRR = 1 mHz ±10%.
   PW = 0.0 ±0.2 and 3.0 ±0.3 V.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
9. Load all outputs as specified on figure 11.
10. Refer to handling considerations in 6.6.

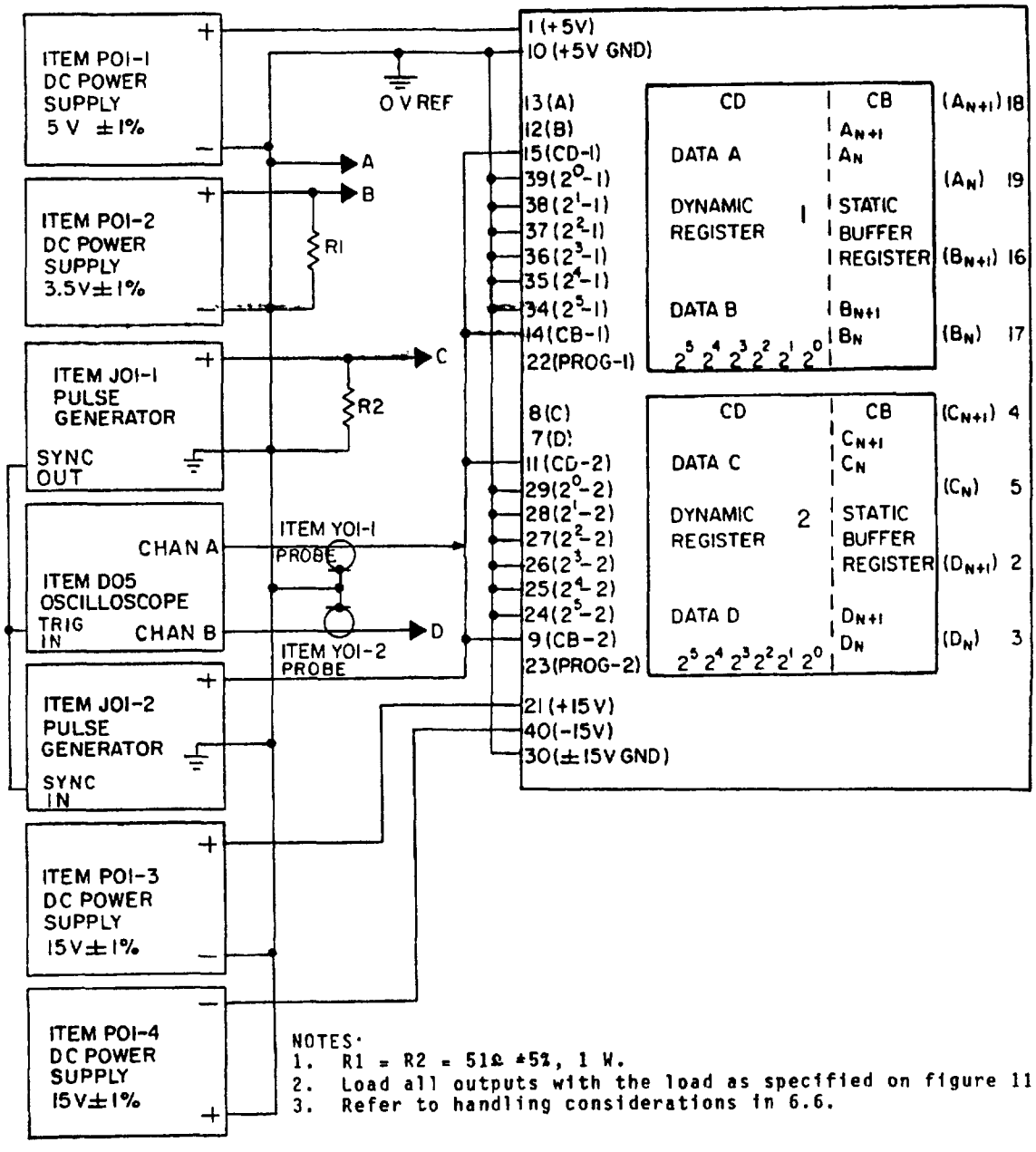
* FIGURE 20. Dynamic shift register operation test circuit for table XVII.

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TABLE XVIII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	A_N	Output crosstalk, low level	13	Open	7,8,12	19	0	400	mV	21,22
S2	A_{N+1}	"	13	"	7,8,12	18	"	"	"	"
S3	B_N	"	12	"	7,8,13	17	"	"	"	"
S4	B_{N+1}	"	12	"	7,8,13	16	"	"	"	"
S5	C_N	"	8	"	7,12,13	5	"	"	"	"
S6	C_{N+1}	"	8	"	7,12,13	4	"	"	"	"
S7	D_N	"	7	"	8,12,13	3	"	"	"	"
S8	D_{N+1}	"	7	"	8,12,13	2	"	"	"	"
S9	A_N	Output crosstalk, high level	Open	13	7,8,12	19	"	"	"	"
S10	A_{N+1}	"	"	13	7,8,12	18	"	"	"	"
S11	B_N	"	"	12	7,8,13	17	"	"	"	"
S12	B_{N+1}	"	"	12	7,8,13	16	"	"	"	"
S13	C_N	"	"	8	7,12,13	5	"	"	"	"
S14	C_{N+1}	"	"	8	7,12,13	4	"	"	"	"
S15	D_N	"	"	7	8,12,13	3	"	"	"	"
S16	D_{N+1}	"	"	7	8,12,13	2	"	"	"	"

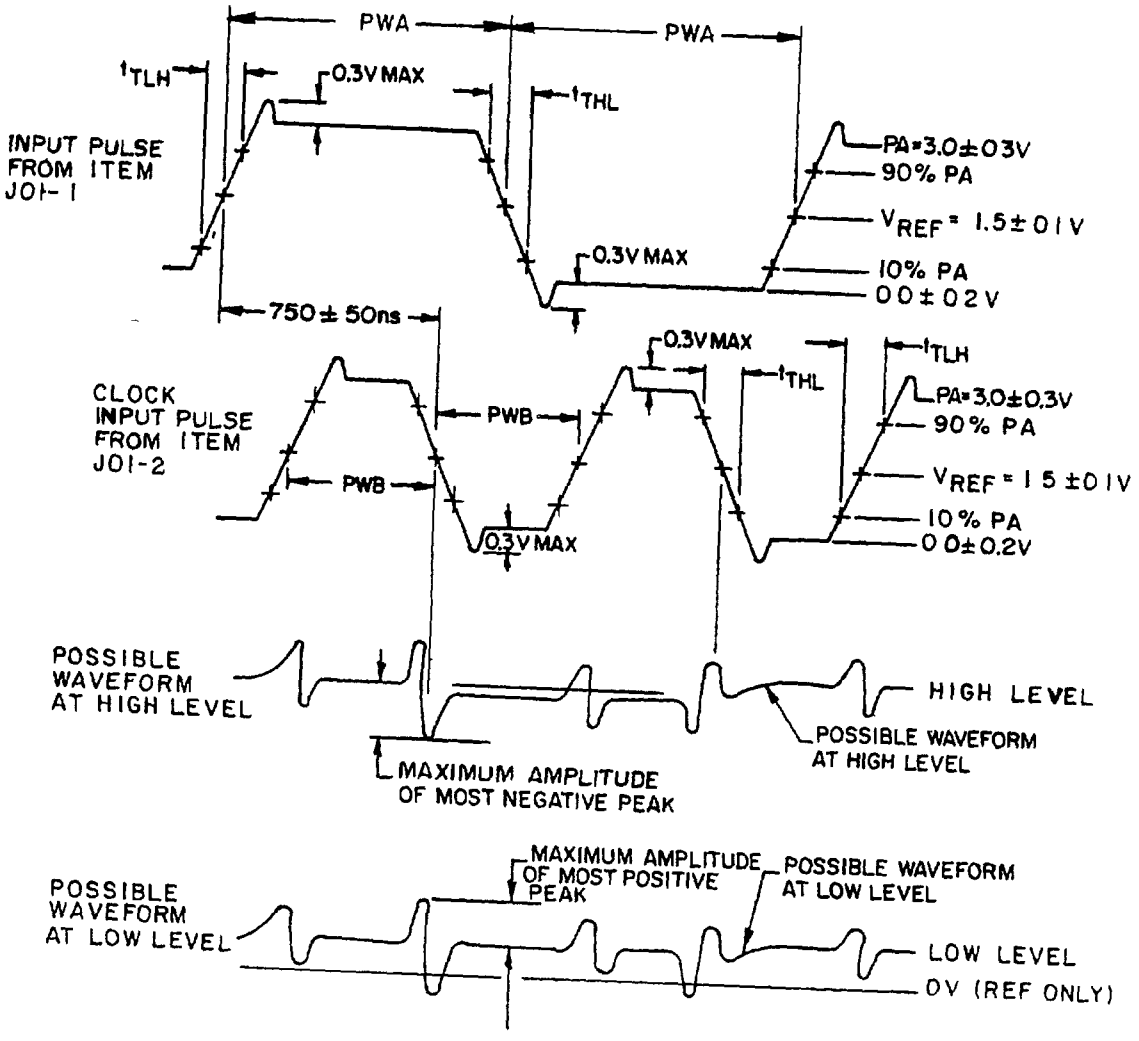
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load all outputs with the load as specified on figure 11.
3. Refer to handling considerations in 6.6.

* FIGURE 21. Output crosstalk (low-level and high-level) test circuit for table XVIII.

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NOTES
1. Item J01-1 input pulse parameters:
   PWB = 500 ±20 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Item J01-2 input pulse parameters:
   PWA = 1,000 ±40 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns
3. All inout voltages are referenced to ground.

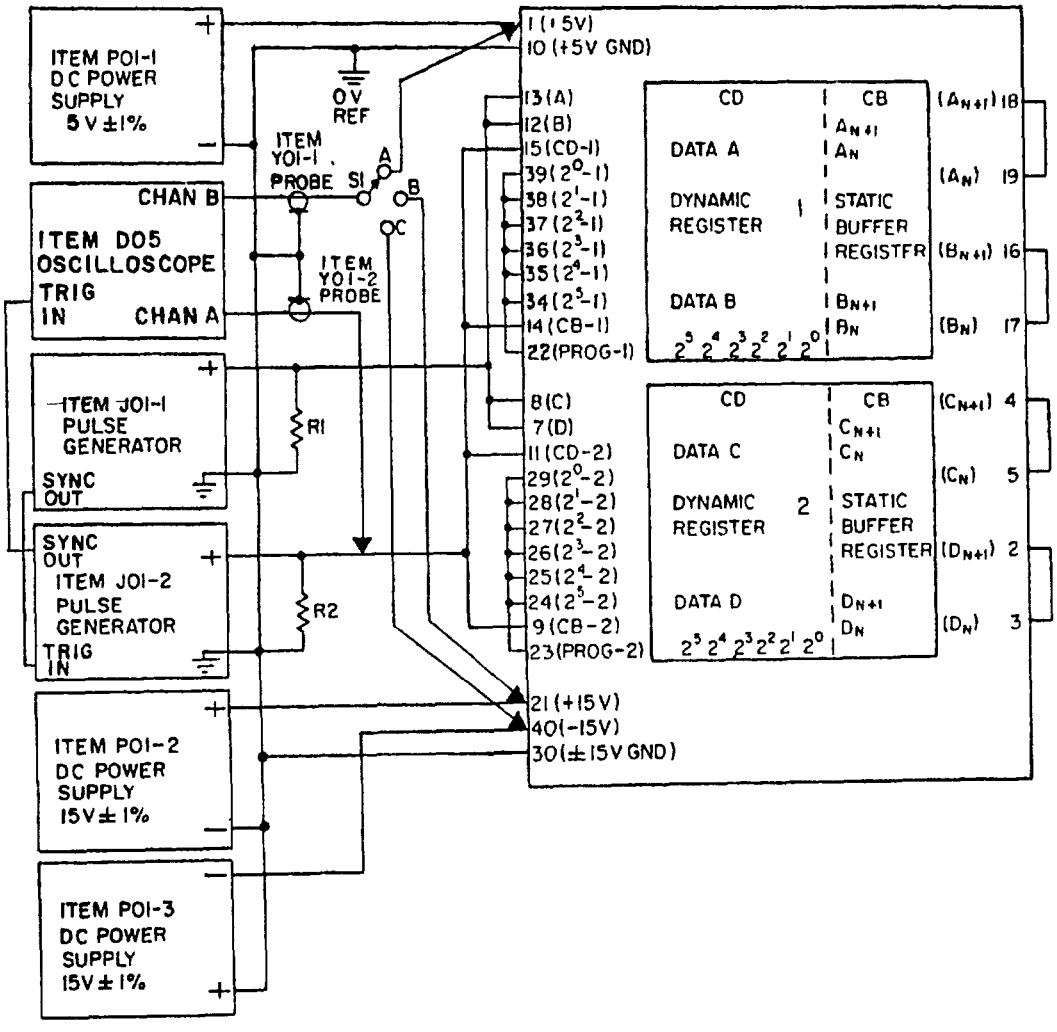
*FIGURE 22 Output crosstalk (low-level and high-level) test circuit waveforms for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
T1	Power supply transient amplitude	0	400	mV	23,24
T2	"	"	"	"	"
T3	"	"	"	"	"

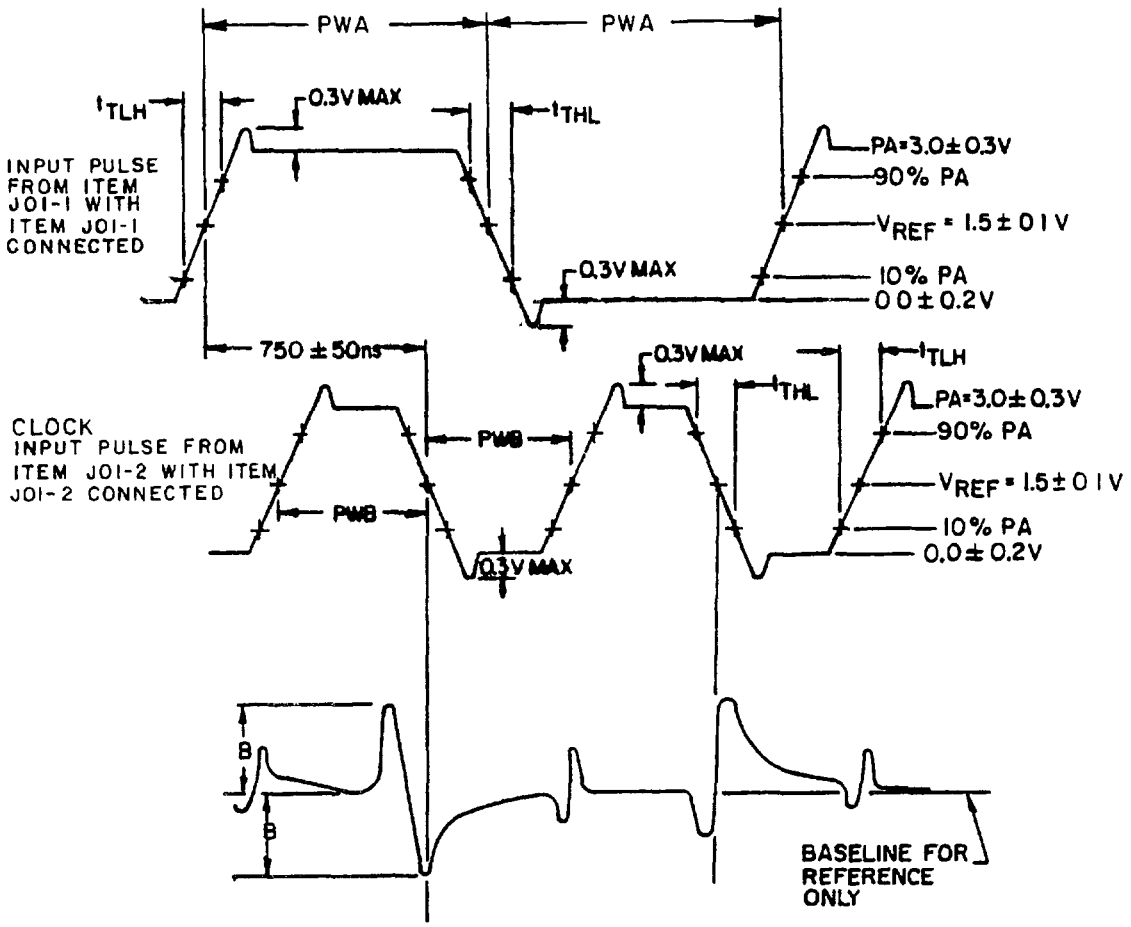
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. S1 = SP3P switch.
3. Place switch S1 in position A for test condition T1, position B for test condition T2, and position C for test condition T3.
4. Refer to handling considerations in 6.6.

* FIGURE 23 Power supply transient amplitude test circuit for table XIX.

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NOTES
1. Item J01-1 input pulse parameters.
   PWA = 1,000 ±40 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Item J01-2 input pulse parameters.
   PWB = 500 ±20 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
3 B = Measurement from the baseline to the most positive peak (T3) and negative (T1 and T2) excursion of the waveform
4 All input voltage levels are referenced to ground.

FIGURE 24 Power supply transient amplitude test circuit waveforms for table XIX.

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*TABLE XX. Power supply filter capacitance test.

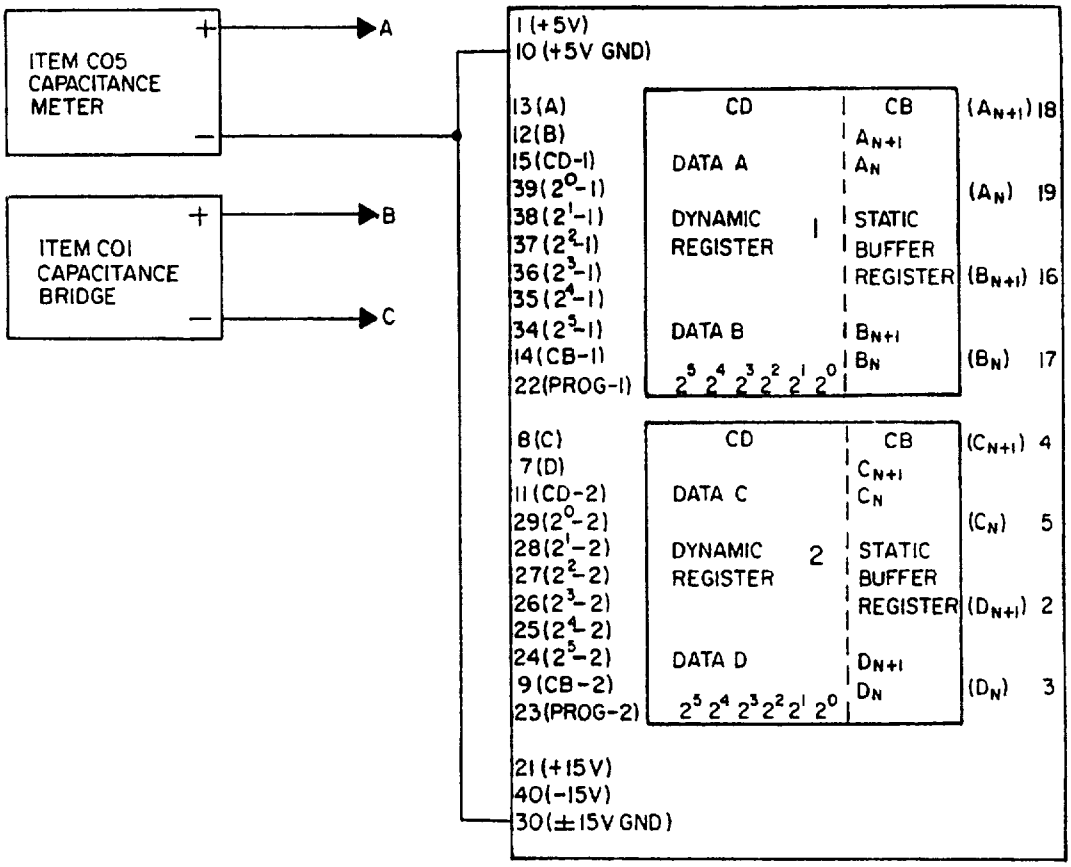
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U	Power supply filter capacitance (5 V)	Open	1	10	1.65	2.75	µF	25

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TABLE XXI. Input capacitance test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
V1	A	Input capacitance	13	Open	Open	2	15	pF	25
V2	B	"	12	"	"	"	"	"	"
V3	CD-1	"	15	"	"	"	"	"	"
V4	2^0-1	"	39	"	"	"	"	"	"
V5	2^1-1	"	38	"	"	"	"	"	"
V6	2^2-1	"	37	"	"	"	"	"	"
V7	2^3-1	"	36	"	"	"	"	"	"
V8	2^4-1	"	35	"	"	"	"	"	"
V9	2^5-1	"	34	"	"	"	"	"	"
V10	CB-1	"	14	"	"	"	"	"	"
V11	C	"	8	"	"	"	"	"	"
V12	D	"	7	"	"	"	"	"	"
V13	CD-2	"	11	"	"	"	"	"	"
V14	2^0-2	"	29	"	"	"	"	"	"
V15	2^1-2	"	28	"	"	"	"	"	"
V16	2^2-2	"	27	"	"	"	"	"	"
V17	2^3-2	"	26	"	"	"	"	"	"
V18	2^4-2	"	25	"	"	"	"	"	"
V19	2^5-2	"	24	"	"	"	"	"	"
V20	CB-2	"	9	"	"	"	"	"	"

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NOTES.
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   a. Frequency = 120 Hz.
   b. Vpk = 0.3 V maximum.
4. Refer to handling considerations in 6.6.

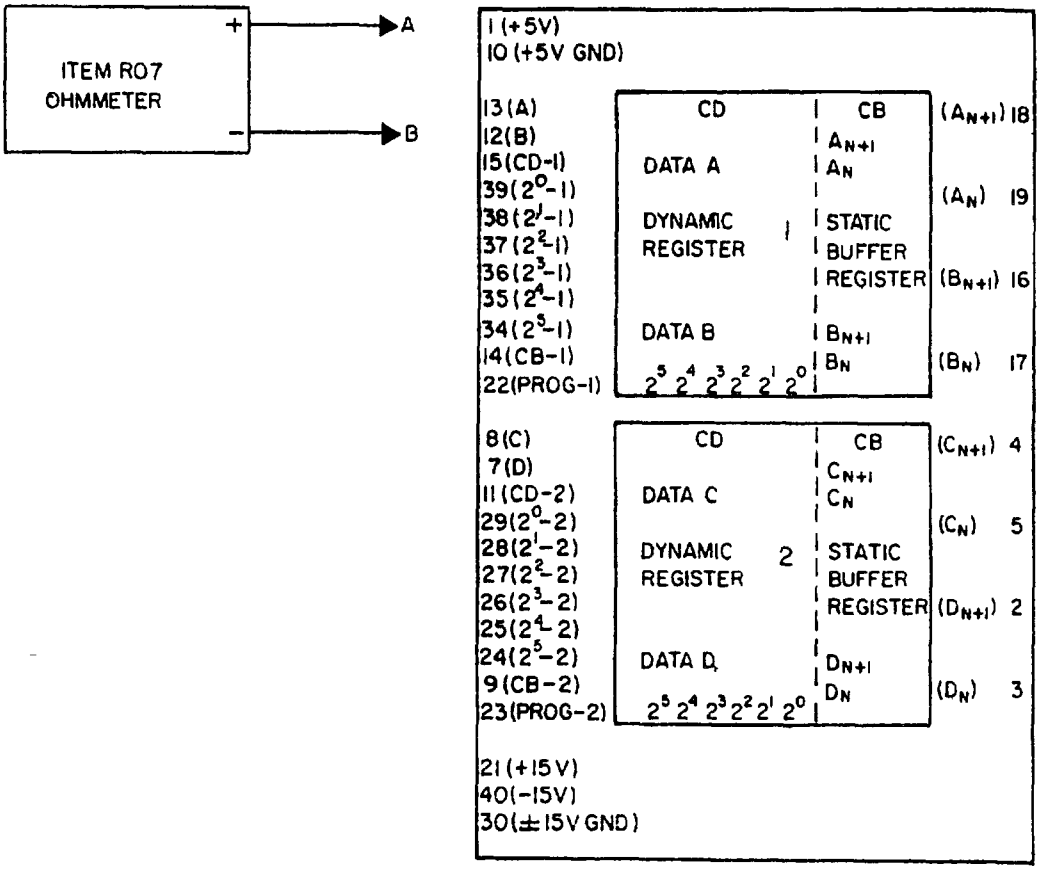
*FIGURE 25. Power supply filter capacitance and input capacitance test circuit for tables XX and XXI.

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TABLE XXII. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	Continuity	22	23	0	1	Ω	26
W2	"	30	31	"	"	"	"
W3	"	30	32	"	"	"	"

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NOTE: Refer to handling considerations in 6.6.

* FIGURE 26. Continuity test circuit for table XXII

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